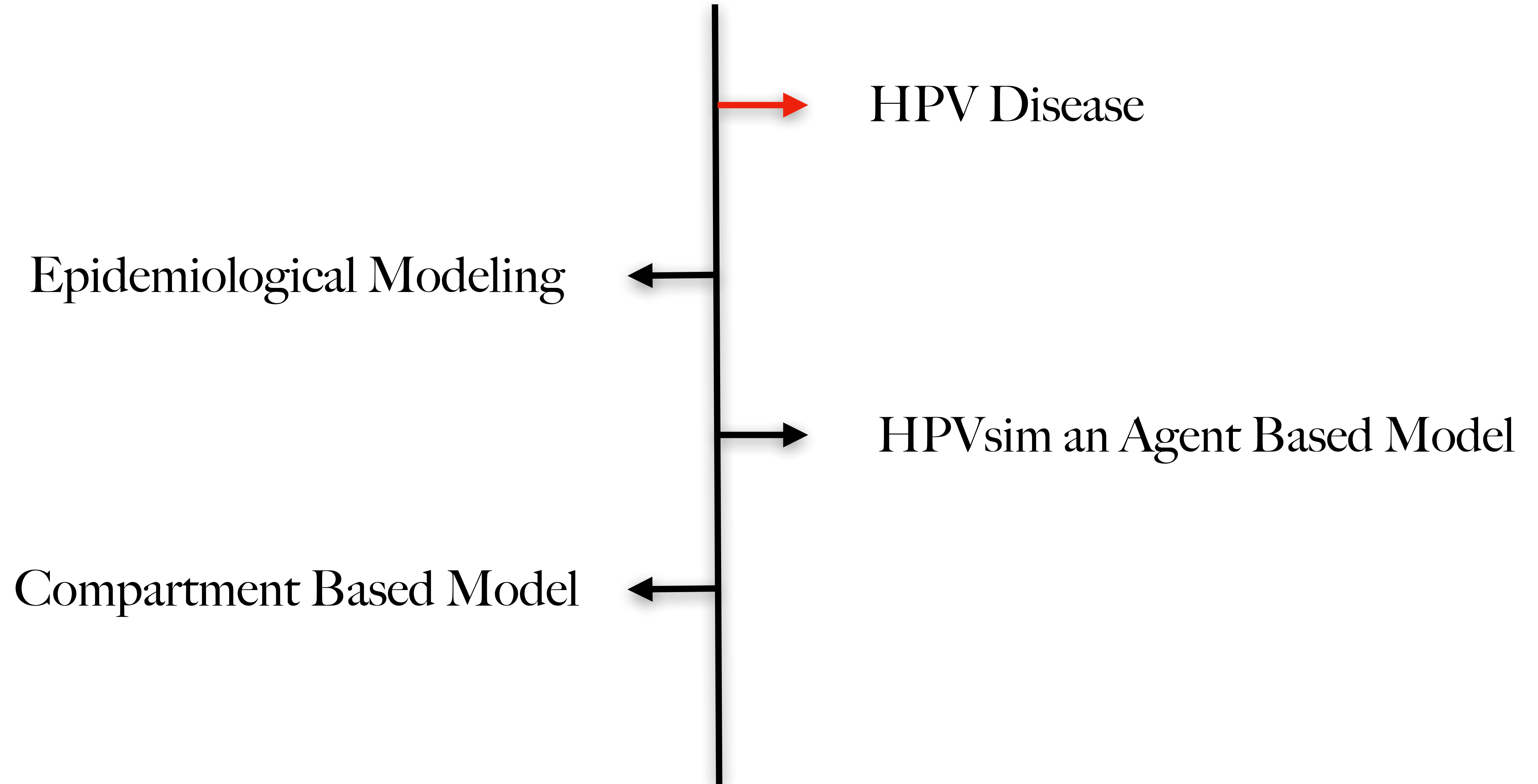


**Integrating Epidemiological Modelling with Optimal Control to
Devise Cost-Effective Intervention Schedules for Human Papilloma
virus Disease Mitigation**

Road map



HPV

Human Papillomavirus Disease

HPV

HPV is a group of more than **200 related viruses**, some of which are spread through close intimate contacts. Those viruses fall into 2 Groups as shown below.

Low Risk

Rarely cause cancer

Few low-risk HPV types can cause warts.

When warts forms in the larynx or respiratory tract, you may have a condition called respiratory papillomatosis, which can cause breathing problems.

High Risk

Cause several types of Cancer

12 high-risk HPV types: HPV 16,18,31,33,35,39,45,51,52,56,58,59

HPV 16 and HPV 18, are responsible for most HPV-related cancers.

What is HPV Disease ?

HPV a common cause for Cervical Cancer CC

- CC is a growth of cells that **starts in the cervix**.
- Over time, if **not destroyed or removed**, the **abnormal cells** may become cancer cells and start to grow and spread more deeply into the cervix and to surrounding areas.
- HPV play a role in causing most cervical cancers.
- When exposed to HPV, the body's immune system typically prevents the virus from doing harm. **In a small percentage of people, however, the virus survives for years.** This contributes to the process that causes some cervical cells to become cancer cells.
- Risk of developing CC be reduced by having **Screening tests** and receiving a **Vaccine** that protects against HPV infection.
- When CC happens, it's often first treated with **surgery** to remove the cancer. Other treatments may include medicines to kill the cancer cells.
- Radiation therapy with powerful energy beams also may be used. Sometimes treatment combines radiation with low-dose chemotherapy.

What is
Cervical
Cancer ?

What causes
Cervical
Cancer ?

If Cervical
Cancer,
What Next ?

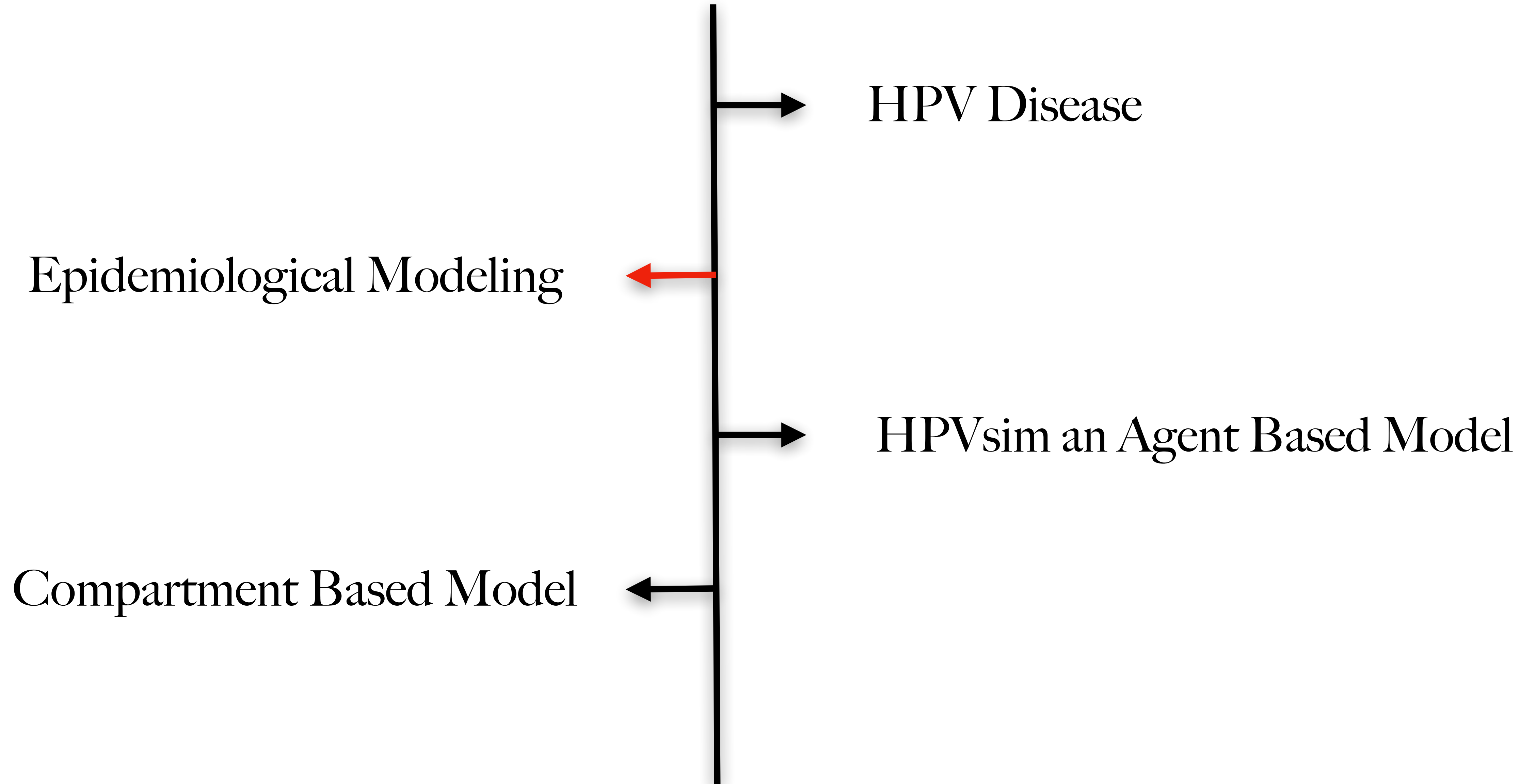
Severity of Cervical Cancer Globally and in India

- Globally, it is estimated that 620K new cancer cases in women and 70 K new cancer cases in men were caused by HPV in 2019.
- CC was the 4th leading cause of cancer and cancer deaths in women in 2022, with some 660K new cases and around 350K deaths worldwide. CC account for over 90 % of HPV-related cancers in women.
- CC is the second most common cancer in Indian women accounting for about 18 % of all cancers occurring in this demographic group.
- Every year, over 120K new cases of this disease are diagnosed out of whom more than 77K cases succumb to death owing to diagnosis during advanced stages of the cancer, bringing the mortality rate to approximately 63 % .

Some Facts
from WHO

Should India
care about
HPV ?

Road map



Epidemiological Modeling

Epidemiological Modeling

If we consider HPV, then it spreads among the population at an Individual Level from one infected person to another.

What if we could track the disease spread, and thus predict a growth of the disease over the population over the time ?

Modeling can help us get quite solid answers to such questions.

We consider here the following types of model for Analysing the HPV disease:

1. Agent Based Model
2. Compartmental Model

Why
model a Disease ?

Agent Based Model vs Compartmental Model

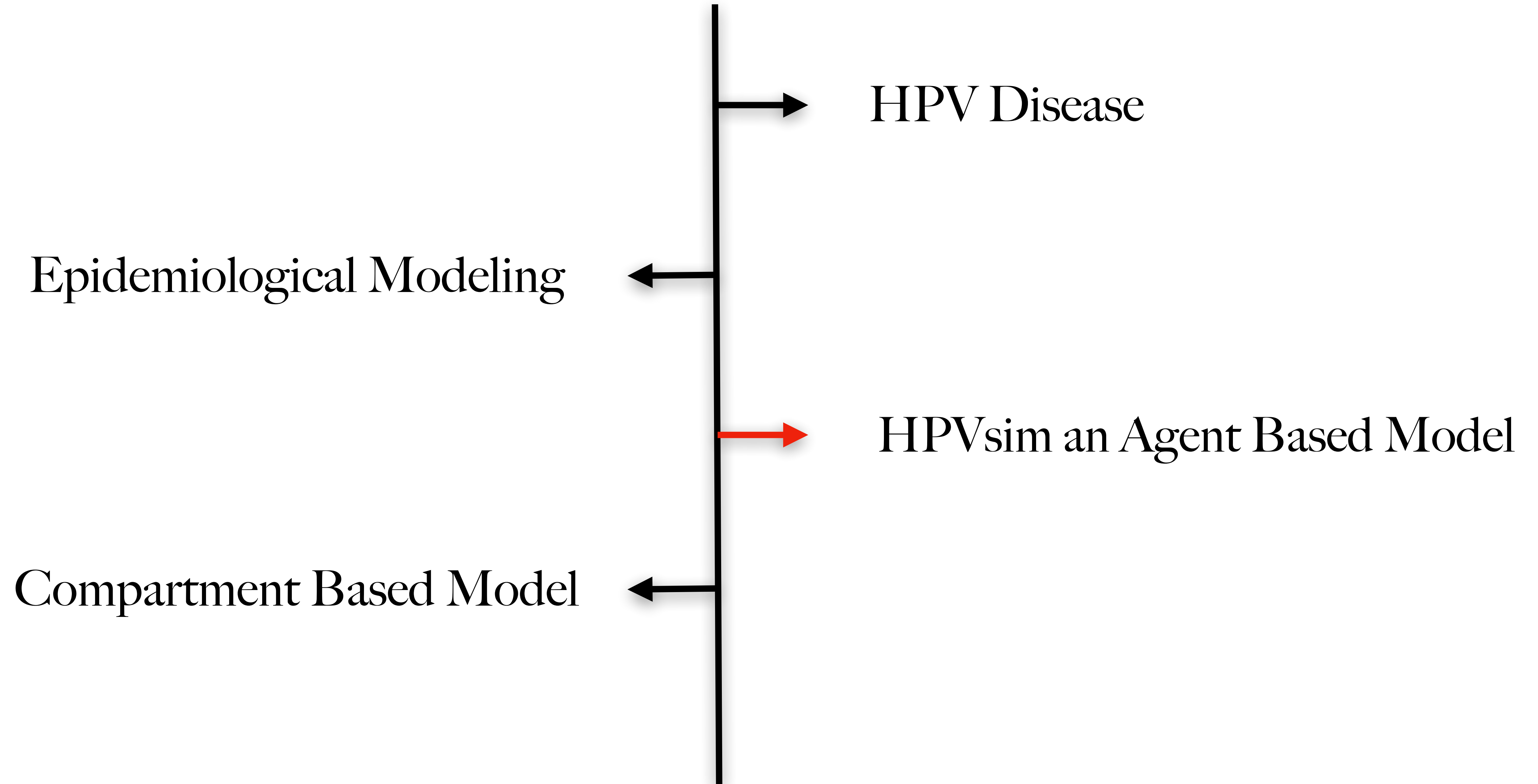
Agent Based Model (ABMs)

- ABMs simulate the behavior of individual entities or "agents" within a system. Each agent typically has its own set of characteristics, behaviors, and rules for interaction with other agents and the environment.
- These models focus on capturing the heterogeneity and interactions between individual entities, allowing for the emergence of complex behaviors and patterns from simple rules.
- ABMs are highly flexible and can represent a wide range of phenomena, but they can also be computationally intensive and require detailed data on individual-level behavior.

Compartmental Models

- Compartmental models, such as the SIR (Susceptible-Infectious-Recovered) model, divide the population into compartments based on their disease status (e.g., susceptible, infectious, recovered).
- These models track the flow of individuals between compartments using a set of differential equations that describe how the number of individuals in each compartment changes over time.
- Compartmental models are relatively simple and computationally efficient, making them suitable for analyzing large populations over longer time scales. However, they may oversimplify individual-level interactions and heterogeneity.

Road map



HPVsim an *Agent Based* Model

HPVsim 101

HPVsim software is an Agent Based Model which needs to be initialised with country specific demographics so as to get results helpful to make decisions tailored to that Country.

To be precise this model can be parameterized with

- Country-specific vital dynamics
 - Structured Mixing networks
 - Co-transmitting HPV genotypes
 - B- and T-cell mediated immunity
 - High-resolution disease natural history
-
- Initial distribution of HPV Genotypes. $hpv16 = 0.4$, $hpv18 = 0.15$, $hi5 = 0.15$, $ohr = 0.3$
 - Average Debut Ages of Males and Females in India.
 - Duration of Partnerships (Marital M+Casual C+One-Off O)
 - Mixing Probability Matrix for M, C and O among different age bins.
 - Efficacy of Contraceptives **0.056**

What is
HPVsim ?

HPVsim
initialisation in
Indian Context

Init 1/2

```
init_genotype_dist['india'] = dict(hpv16=0.4, hpv18=0.15, hi5=0.15, ohr=0.3)

debut['india'] = dict(
    f=dict(dist='normal', par1=17.8, par2=2.0), # No data, assumption
    m=dict(dist='normal', par1=17.0, par2=2.0)) # No data, assumption

partners['india'] = dict(
    m=dict(dist='poisson', par1=0.05),
    c=dict(dist='poisson', par1=0.2),
    o=dict(dist='poisson', par1=4),
)

dur_pship['india'] = dict(m=dict(dist='normal_pos', par1=20, par2=3),
                           c=dict(dist='normal_pos', par1=1, par2=1),
                           o=dict(dist='normal_pos', par1=0.5, par2=1))
```

```
layer_probs['india'] = dict(
    m=np.array([
        [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75],
        [0, 0, 0, 0.05, 0.25, 0.35, 0.3, 0.2, 0.1, 0.05, 0.01, 0.005, 0.005, 0.001, 0.001, 0.001], # Share of females of each age who are married
        [0, 0, 0, 0.01, 0.25, 0.35, 0.3, 0.2, 0.1, 0.05, 0.01, 0.005, 0.005, 0.001, 0.001, 0.001]] # Share of males of each age who are married
    ),
    c=np.array([
        [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75],
        [0, 0, 0.10, 0.5, 0.6, 0.6, 0.6, 0.1, 0.05, 0.01, 0.01, 0.01, 0.01, 0.005, 0.001, 0.001], # Share of females of each age having casual relationships
        [0, 0, 0.05, 0.5, 0.6, 0.6, 0.6, 0.5, 0.4, 0.4, 0.3, 0.2, 0.15, 0.1, 0.01, 0.01]], # Share of males of each age having casual relationships
    ),
    o=np.array([
        [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75],
        [0, 0, 0.01, 0.01, 0.01, 0.02, 0.02, 0.02, 0.01, 0.01, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001], # Share of females of each age having one-off relationships
        [0, 0, 0.01, 0.01, 0.01, 0.02, 0.03, 0.02, 0.02, 0.02, 0.002, 0.002, 0.001, 0.001, 0.001, 0.001]], # Share of males of each age having one-off relationships
    ),
)
```

Init 2/2

```
m=np.array([
#   0,  5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[10, 0, 0, .1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[15, 0, 0, .1, .1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[20, 0, 0, .1, .1, .1, .1, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[25, 0, 0, .5, .1, .5, .1, .1, 0, 0, 0, 0, 0, 0, 0, 0],
[30, 0, 0, 1, .5, .5, .5, .5, .1, 0, 0, 0, 0, 0, 0, 0],
[35, 0, 0, .5, 1, 1, .5, 1, 1, .5, 0, 0, 0, 0, 0, 0],
[40, 0, 0, 0, .5, 1, 1, 1, 1, 1, .5, 0, 0, 0, 0, 0],
[45, 0, 0, 0, 0, .1, 1, 1, 2, 1, 1, .5, 0, 0, 0, 0],
[50, 0, 0, 0, 0, 0, .1, 1, 1, 1, 1, 2, .5, 0, 0, 0],
[55, 0, 0, 0, 0, 0, 0, .1, 1, 1, 1, 1, 2, .5, 0, 0],
[60, 0, 0, 0, 0, 0, 0, 0, .1, .5, 1, 1, 1, 2, .5, 0],
[65, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 2, .5],
[70, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, .5],
[75, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1],
]),
```

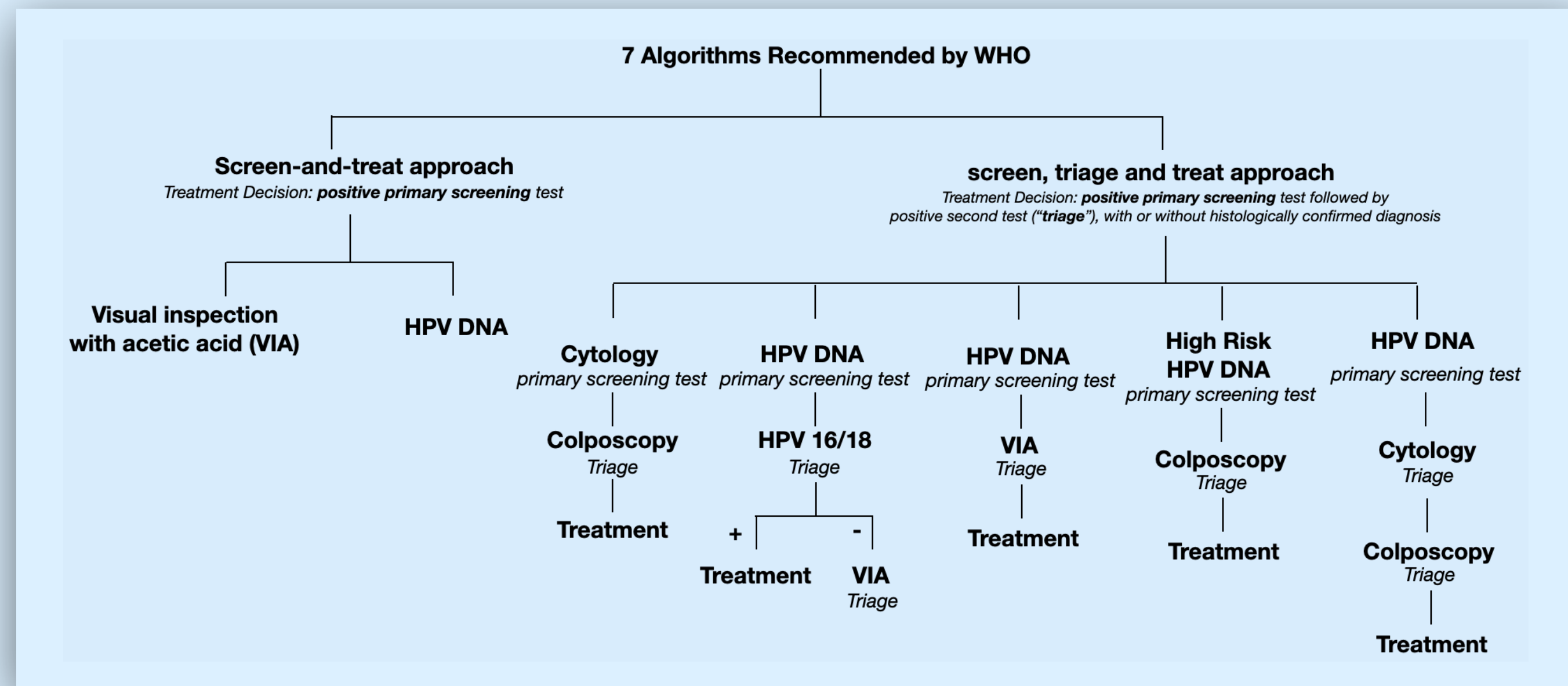
```
c=np.array([
#   0,  5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[10, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[15, 0, 0, 1, 1, 1, 1.0, 0.5, 0, 0, 0, 0, 0, 0, 0, 0],
[20, 0, 0, .5, 1, 1, 1, 1.0, 0.5, 0, 0, 0, 0, 0, 0, 0],
[25, 0, 0, 0, 1, 1, 1, 1.0, 1.0, 0.5, 0, 0, 0, 0, 0, 0],
[30, 0, 0, 0, .5, 1, 1, 1, .5, 0.5, 0.5, 0, 0, 0, 0, 0],
[35, 0, 0, 0, .5, 1, 1, 1, 1, .5, 0.5, 0.5, 0, 0, 0, 0],
[40, 0, 0, 0, .5, .5, 1, 1, 1, 1, .5, 0.5, 0, 0, 0, 0],
[45, 0, 0, 0, 0.5, 0.5, 1, 1, 1, 1, 1, .5, 0, 0, 0, 0],
[50, 0, 0, 0, 0.5, 0.5, 0.5, 1, 1, 1, 1, 1, .5, 0, 0, 0],
[55, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0, 0],
[60, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0],
[65, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 2, .5],
[70, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, .5],
[75, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1],
]),
```

```
o=np.array([
#   0,  5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75
[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[10, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[15, 0, 0, 1, 1, .5, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[20, 0, 0, .5, 1, 1, .5, 0, 0, 0, 0, 0, 0, 0, 0, 0],
[25, 0, 0, 0, 1, 1, 1, .5, 0, 0, 0, 0, 0, 0, 0, 0],
[30, 0, 0, 0, 0, 1, 1, 1, .5, 0, 0, 0, 0, 0, 0, 0],
[35, 0, 0, 0, 0, 1, 1, 1, 1, .5, 0, 0, 0, 0, 0, 0],
[40, 0, 0, 0, 0, 0, 1, 1, 1, 1, .5, 0, 0, 0, 0, 0],
[45, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0, 0, 0, 0],
[50, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0, 0, 0],
[55, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0, 0],
[60, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, .5, 0],
[65, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 2, .5],
[70, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, .5],
[75, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1],
]),
```

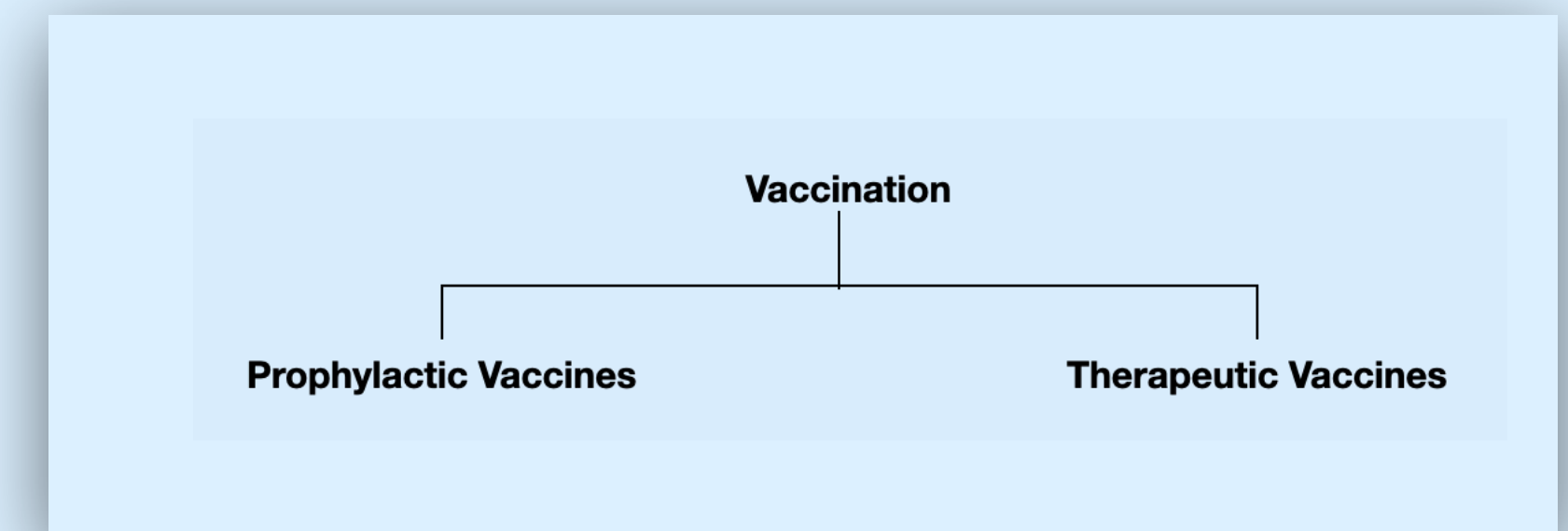
Usability of HPVsim: Designing Intervention Schedules

- HPVsim can be used to test out various intervention schedules i.e. a Methodology to curb the growth of the disease and how do they work over period of time if applied over a probabilistically simulated population growth.
- Interventions: Screening and Treatment Strategies + Vaccination Strategies.

Screening and Treatment Strategies



Vaccination Strategies



- Apart from above a custom strategy along with a custom schedule could also be developed.

Experimentation with HPVsim to devise a Good Intervention Strategy

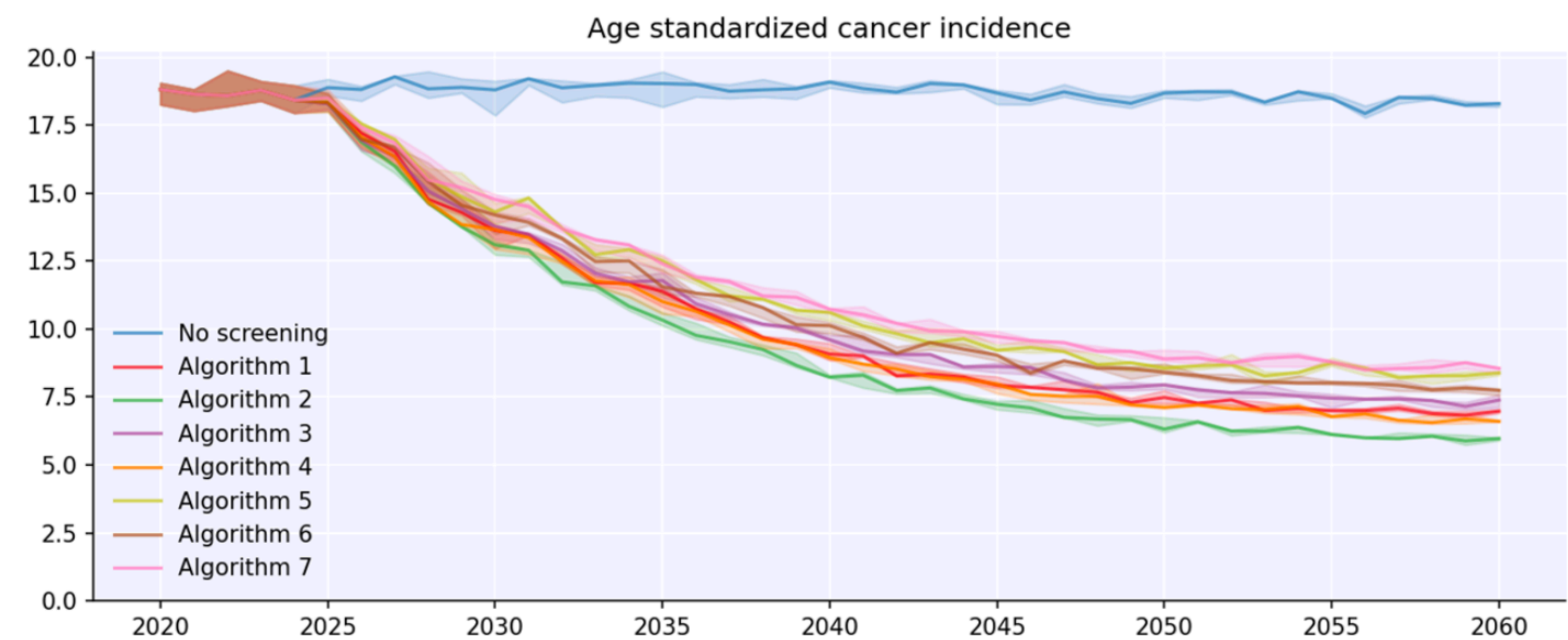
Greedy Analysis

- (i) No Intervention
- (ii) Only Performing Screening and Treatment (S&T)
 - Comparing all 7 S&T strategies chalked out by WHO
- (iii) Only Performing Vaccination (V)
 - (a) Male Only Vaccination
 - (b) Female Only Vaccination
 - (c) Gender Neutral Vaccination
 - Identifying Good M:F ratio
 - Identifying right Age Bracket for Vaccinating

Non Greedy Analysis

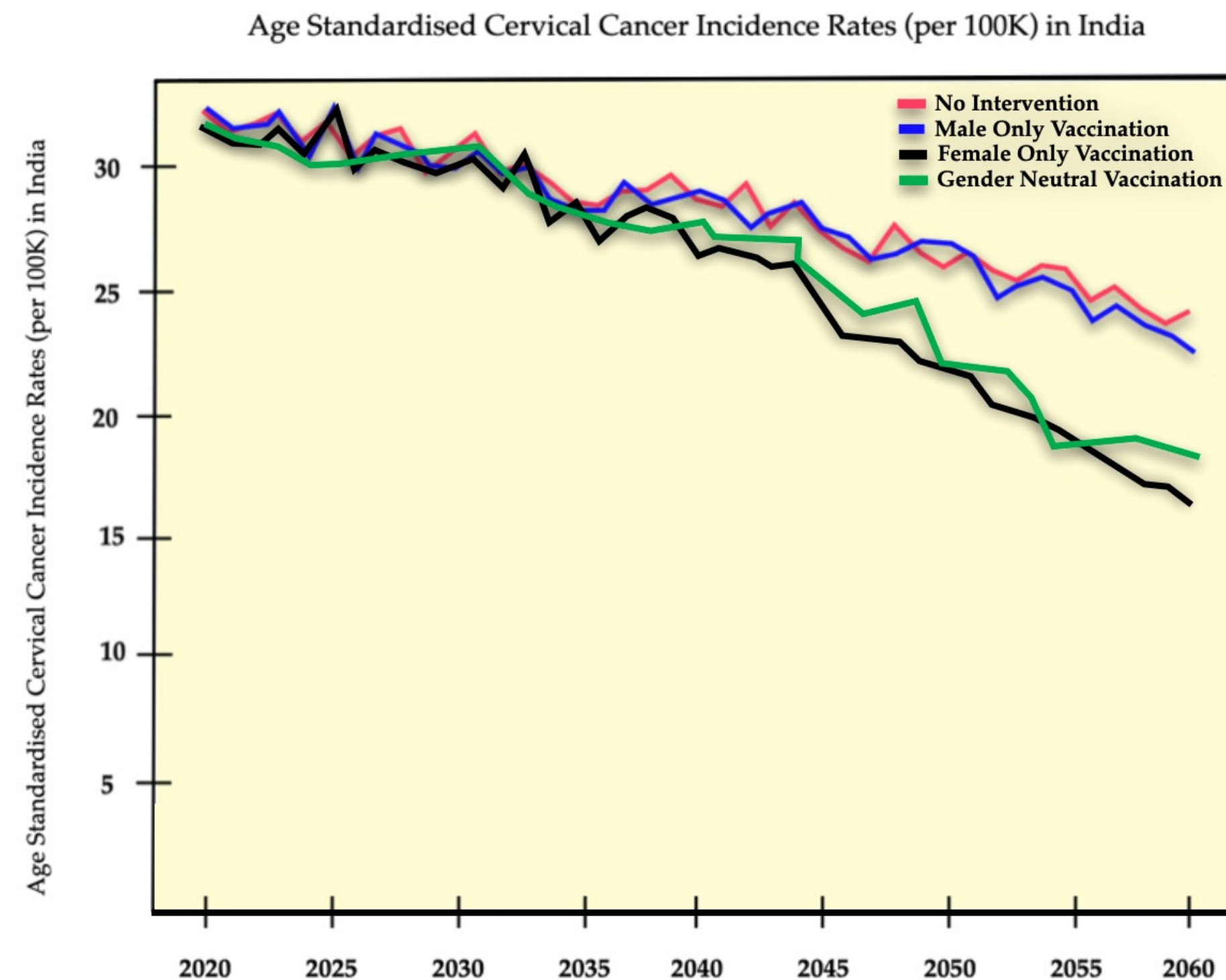
- (i) Combining S&T and V
- (ii) Anomaly Detection in Time Series Data (ECG Data)
- (iii) Anomaly Detection in Networked Data (Packet Transfer)

Greedy Analysis: Algorithm 2 turns out to be the Best balancing Cost and Efficacy Both



Product	Cost (2022 USD)
VIA	\$13
HPV DNA test	\$20
HPV DNA genotype test	\$30
Cytology	\$20
Colposcopy	\$20
Thermal ablation	\$5

Greedy Analysis: Gender Neutral Vaccination should be Promoted but “Sparingly”

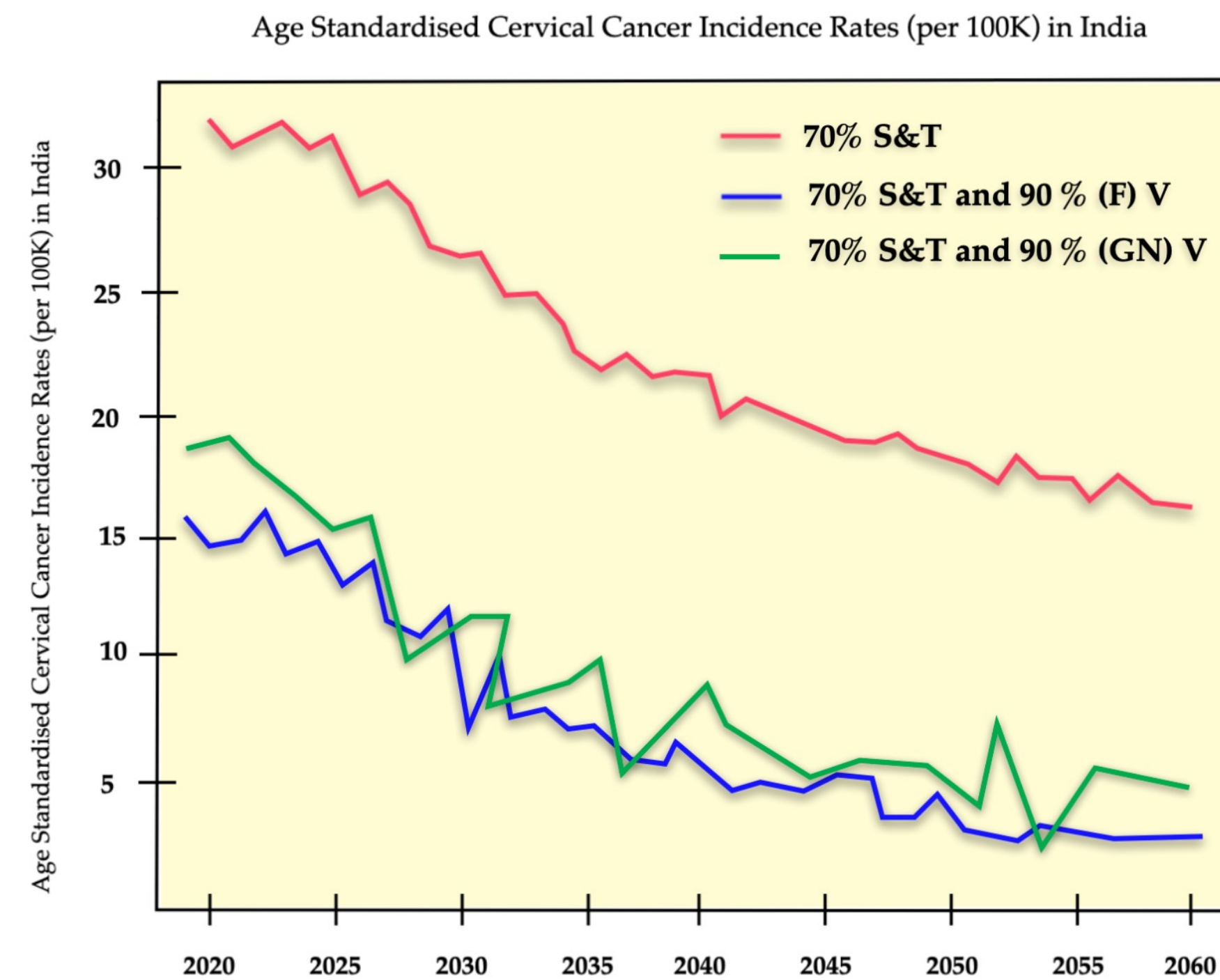
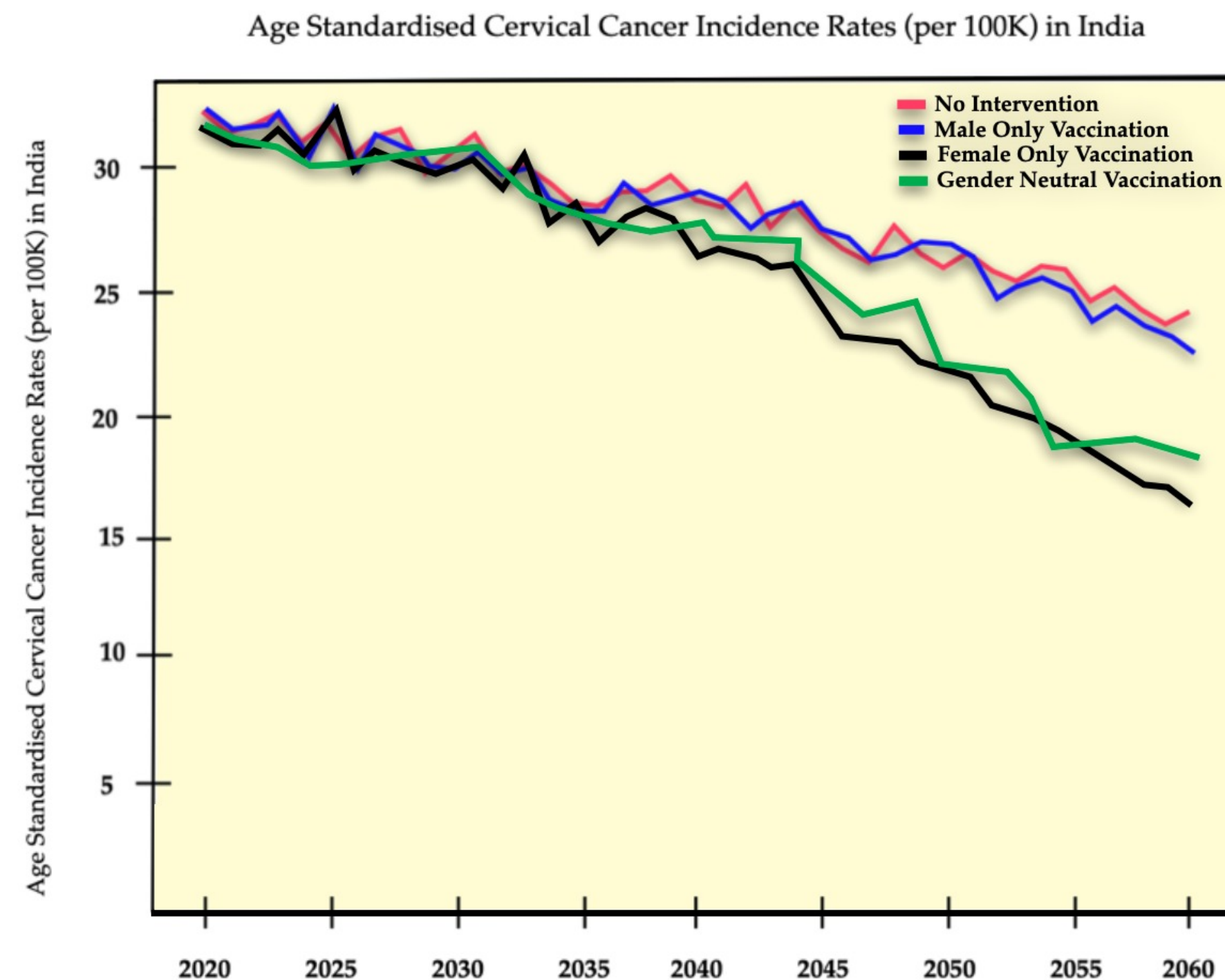


Intervention
Strategy:
Only
Vaccination

Default % for V for
Female as 90%

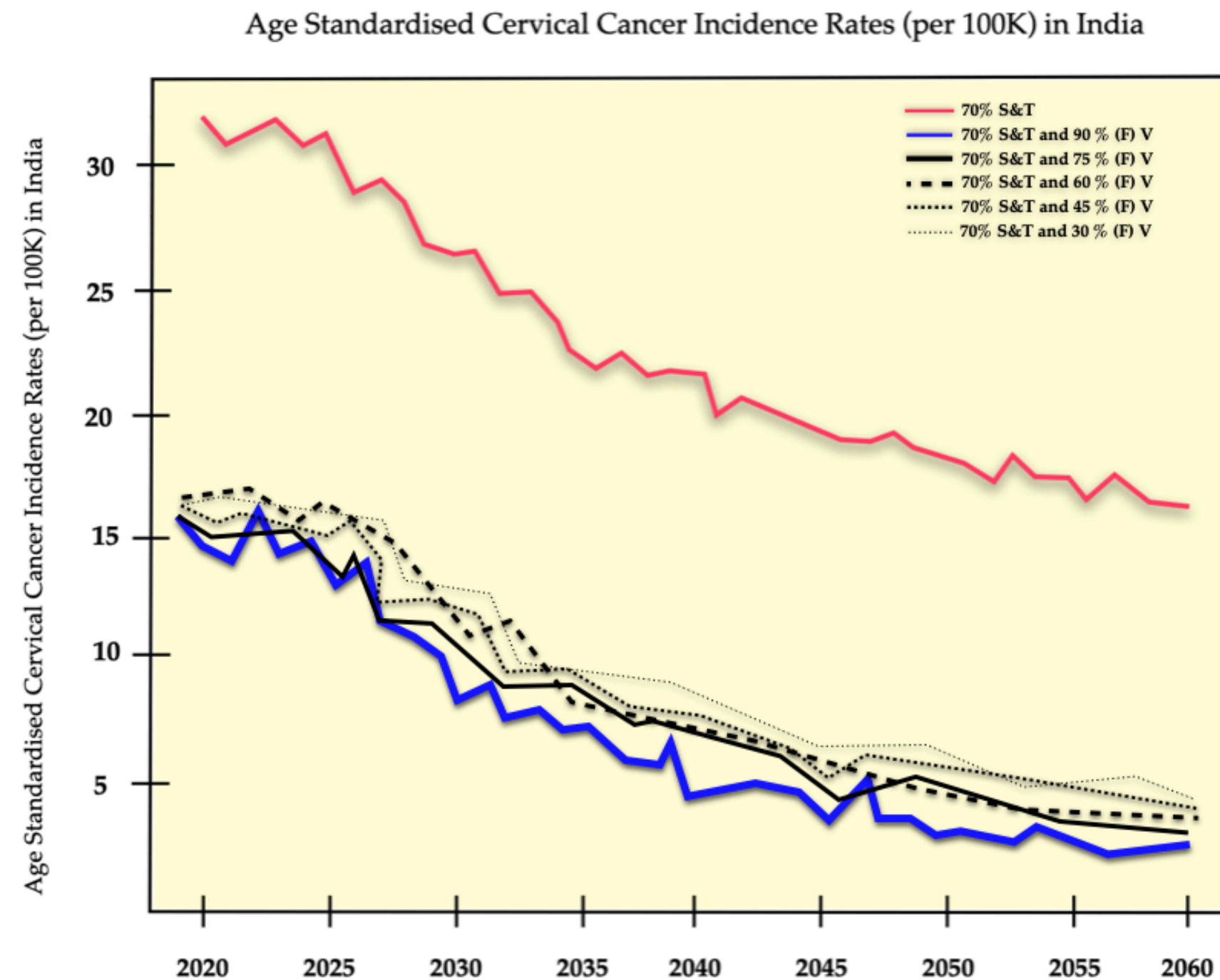
Female Vaccination should be definitely the final solution to handle HPV-led CC, as Vaccinating Males alone show no significant improvement over the case when Not Vaccinated at all. But to reduce the “Female-Only” stigma, Males can be Vaccinated but then it should be done “Sparingly”.

Non Greedy Analysis: S&T and V should be Combined in a Good Intervention Schedule

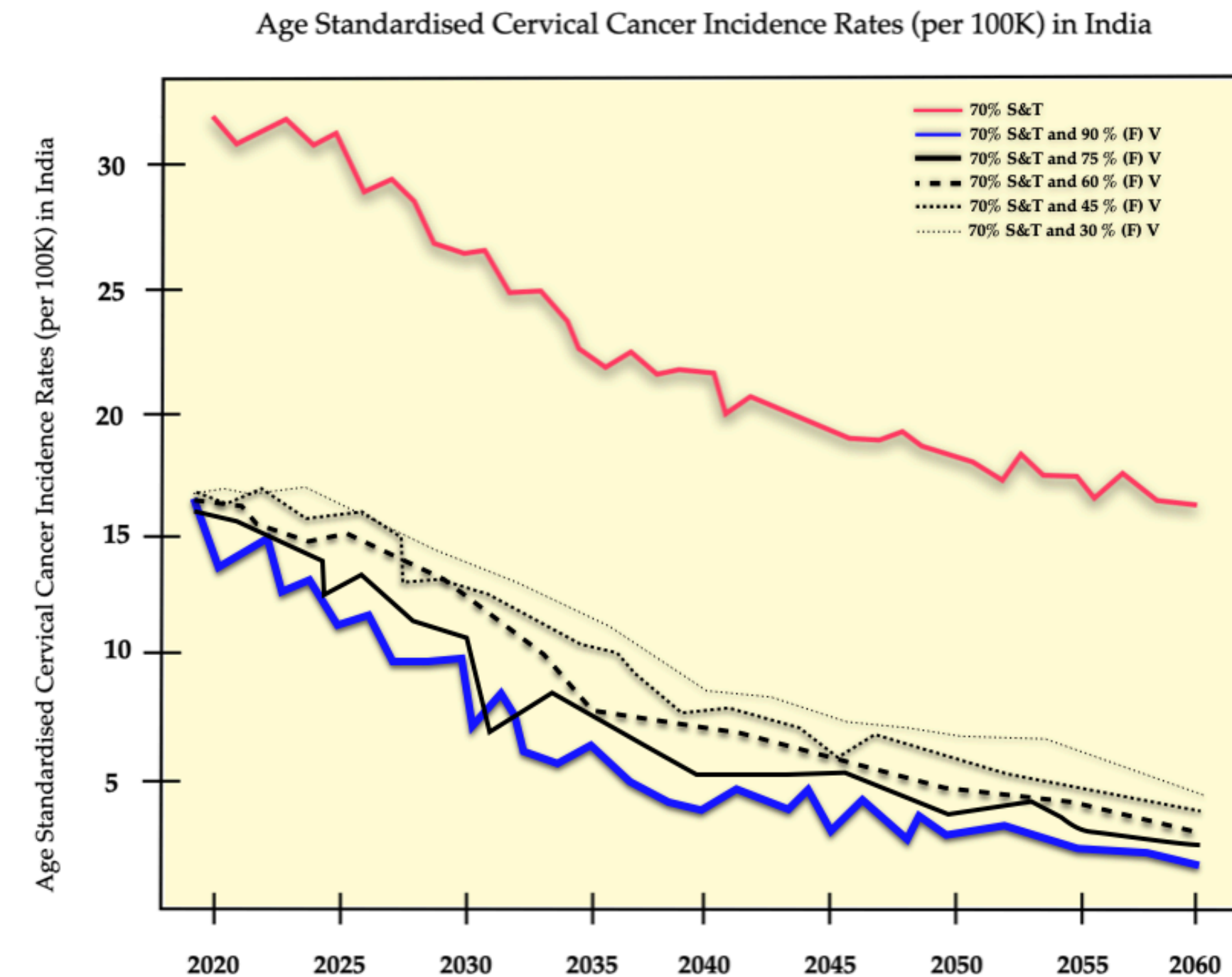


The Only "S&T" intervention schedule (Right Side Red Line) shows a Concave (faster) decline in the initial years whereas the Only "V" intervention schedule (Left Side), shows a Convex (slower) decline in the initial years, which resonates quite well with the fact that S&T helps secure the present while V secures the future. The integration of both intervention schedules, designed to balance future preparedness with present stability, demonstrates an enhanced overall impact, as evidenced by the observed declines (Right Side Plot Blue + Green Lines). Consequently, an effective strategy for a nation such as India would entail maintaining a robust S&T Mechanism consistently operational, while strategically determining the commencement and pace of the "V" intervention schedule to fulfill specific needs as they arise.

Greedy Analysis : Vaccination Age Bracket Should be Well-Thought Off



Vaccination Age Range 9-14



Vaccination Age Range 13-18

Vaccinating only girls aged 9-14 may be a myopic decision and instead, this "suggested" range itself needs to be well thought off. One factual argument could be made that the mixing ages in the Western countries are way below what we have in India (in Ours on avg it's about 20, whereas in the West its about 15) hence an age range of 9-14 is obviously seems and "under-estimate" for India. Ideally it should be something higher extending till at least age of 18 or 19. This fact also resonates quite well in the right-hand side plot shown above as contrasted with the left-hand side plot where I am targeting different age bands.

So Combined S&T, V is Good. What Next ?

Identifying the “Right” Splits

To find Optimal percentages α and β

Which will lead to following Course of Action

$$\alpha \cdot (S \& T) + \beta \cdot (V) \quad \beta \ll \alpha$$

Where we will want the following to hold

1. CIR must decline by K %

$$2. \alpha \cdot (\text{Cost of } S \& T) + \beta \cdot (\text{Cost of } V) \leq \text{Budget } B$$

$$3. V = m \cdot (V_{maleVacc}) + f \cdot (V_{femaleVacc}) \quad f \gg m$$

So in this

We would like to fix K

And obtain Corresponding

α and β

Identifying the “Right” Splits

Considering the above Objective

Next Course of Action

To build a Model, which will try to learn the pattern of CIR in Nations which have Good amount of work already done in this field.

Based on that we would be interested in Divergence Factor δ

$$\alpha \cdot (S \& T) + \beta \cdot (V) \implies K \% \text{ decline CIR} \quad \text{US}$$

$$\alpha_{delta} = \alpha \cdot \delta \quad \beta_{delta} = \beta \cdot \delta$$

$$\alpha_{\delta} \cdot (S \& T) + \beta_{\delta} \cdot (V) \implies K \% \text{ decline CIR} \quad \text{INDIA}$$

That's how we could implement Transfer Learning in Our Context

Coming back to the 2 Non Greedy Tasks

Greedy Analysis

- (i) No Intervention
- (ii) Only Performing Screening and Treatment (S&T)
 - Comparing all 7 S&T strategies chalked out by WHO
- (iii) Only Performing Vaccination (V)
 - (a) Male Only Vaccination
 - (b) Female Only Vaccination
 - (c) Gender Neutral Vaccination
 - Identifying Good M:F ratio
 - Identifying right Age Bracket for Vaccinating

Non Greedy Analysis



- (i) Combining S&T and V
- (ii) Anomaly Detection in Time Series Data (ECG Data)
- (iii) Anomaly Detection in Networked Data (Packet Transfer)

So these 2 are explored branches and could definitely be taken further as an advancement to the existing work done.

Anomaly Detection in Time Series Data (ECG Data)

Building a Model to detect Deviation in the TimeSeries

Case Study: ECG Brain Signal Analysis

The dataset contains 5,000 Time Series examples (obtained with ECG) with 140 timesteps. Each sequence corresponds to a single heartbeat from a single patient with congestive heart failure.

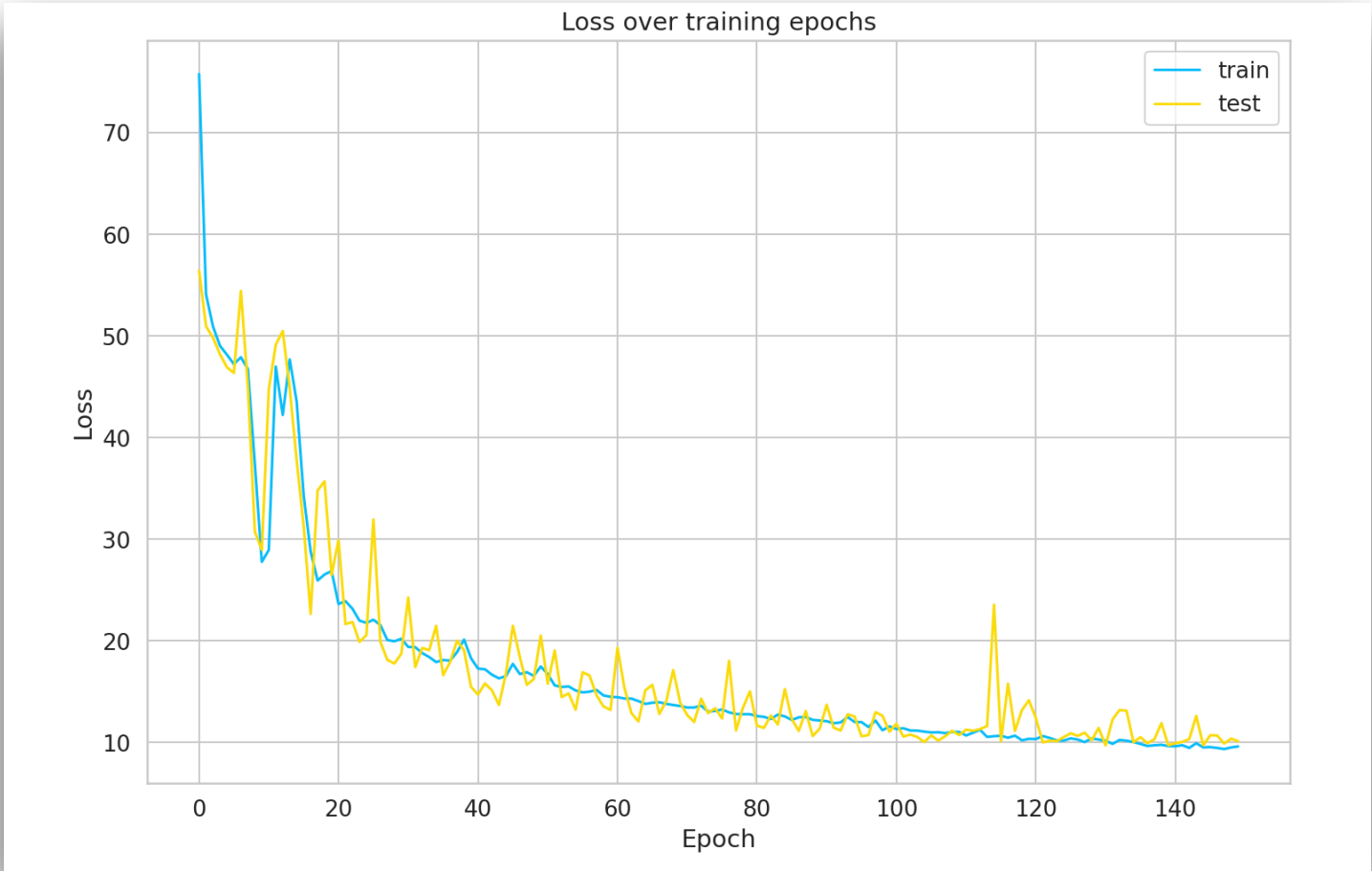
An electrocardiogram (ECG or EKG) is a test that checks how your heart is functioning by measuring the electrical activity of the heart. With each heart beat, an electrical impulse (or wave) travels through your heart. This wave causes the muscle to squeeze and pump blood from the heart. Source

We have 5 types of hearbeats (classes):

1. Normal (N)
2. R-on-T Premature Ventricular Contraction (R-on-T PVC).
3. Premature Ventricular Contraction (PVC)
4. Supra-ventricular Premature or Ectopic Beat (SP or EB)
5. Unclassified Beat (UB).

Now here the idea was to Build a Model which will flag an Anomalous behaviour found in a Timeseries of a Normal Case VS an Abnormal Case.

Model Details and Results



140x1

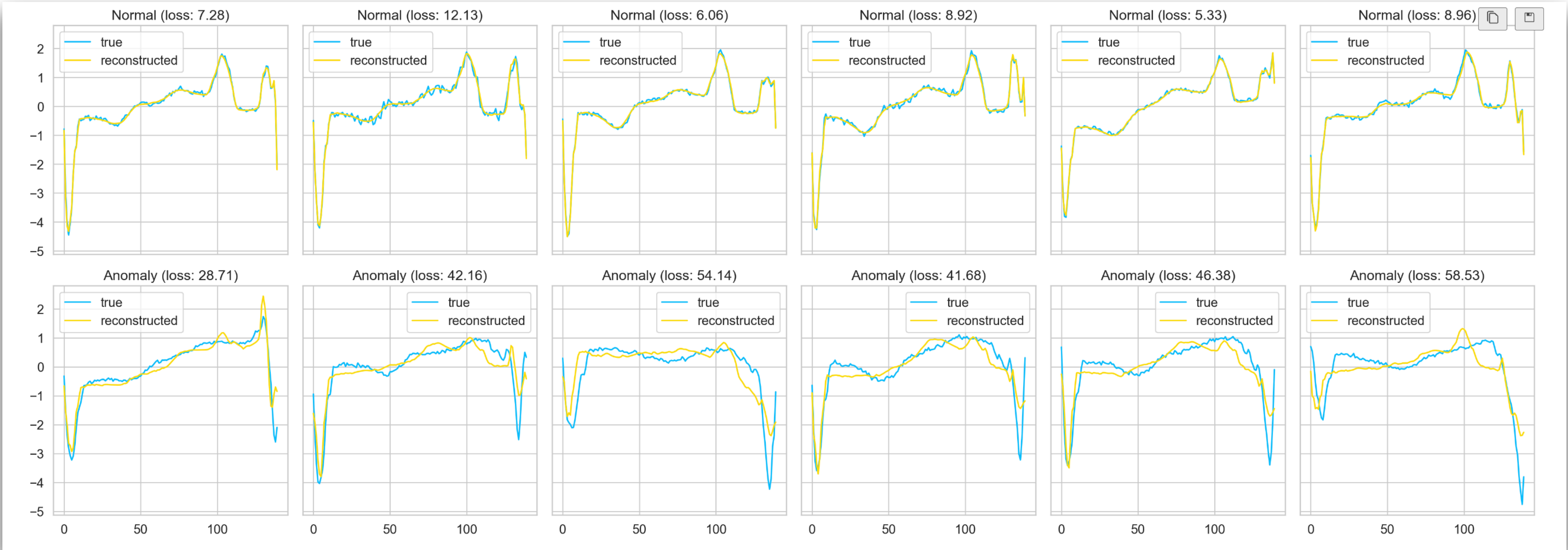
LSTM1 140x128

LSTM2 140x64

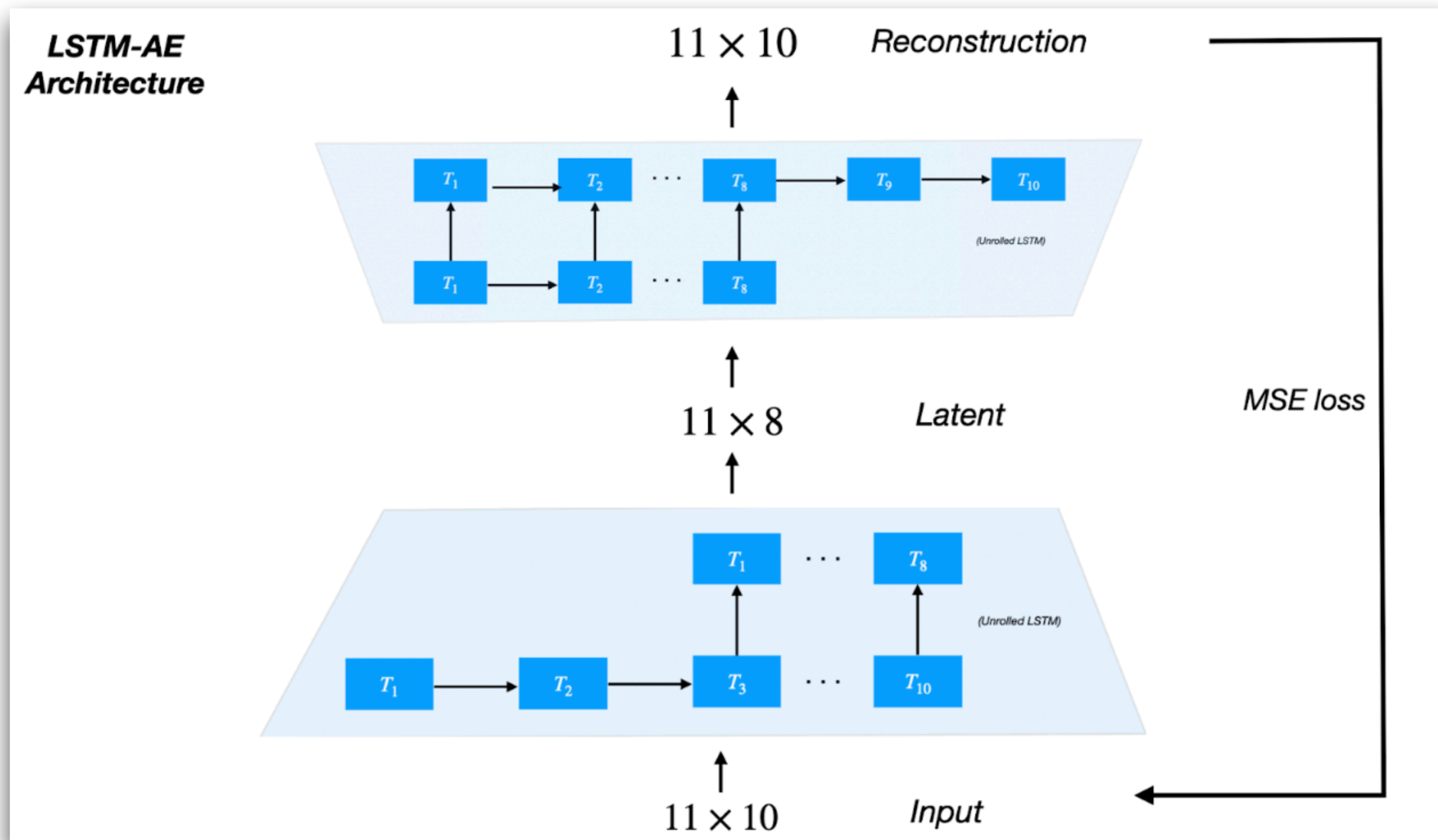
Embedding 64x1

LSTM2 140x64

LSTM1 140x128



Timeseries Anomaly Detection Between US CIR and India CIR



MSE loss for training dataset over US, comes out as 0.15 and with another CIR Timeseries Data from India it comes out as 1.076

Interpretation of the Reconstruction Loss using the LSTM based Auto-Encoder Model was Indeed Challenging.

Identifying Anomalous behaviour spotted in the Network

**NetDiffusion: Network Data Augmentation Through Protocol-
Constrained Traffic Generation**

Over View of the Model

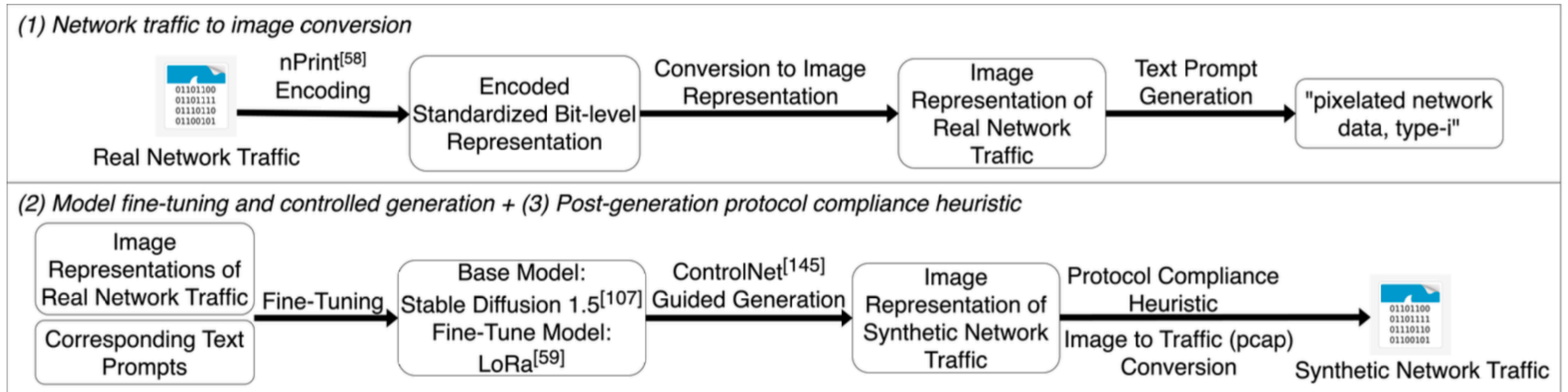
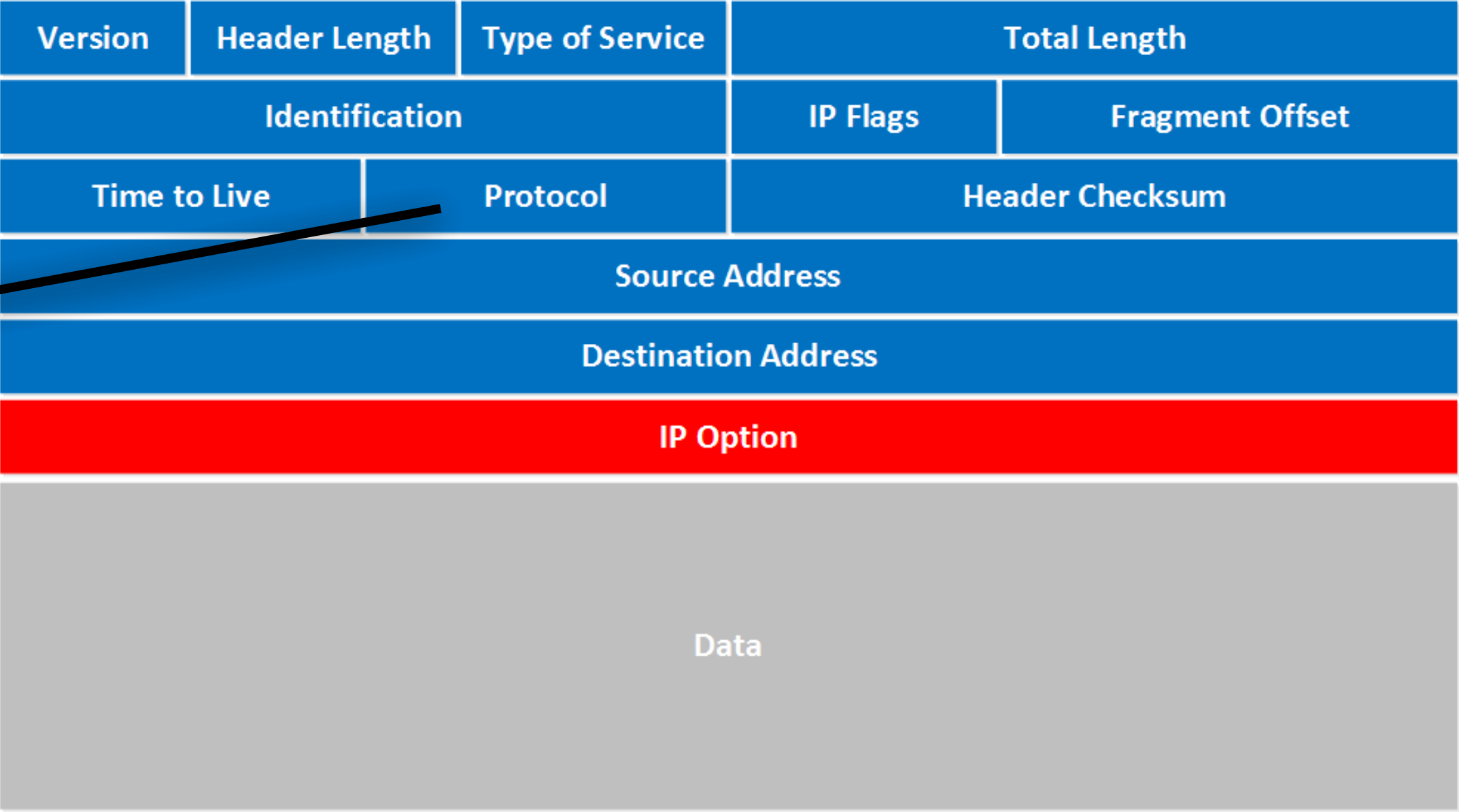


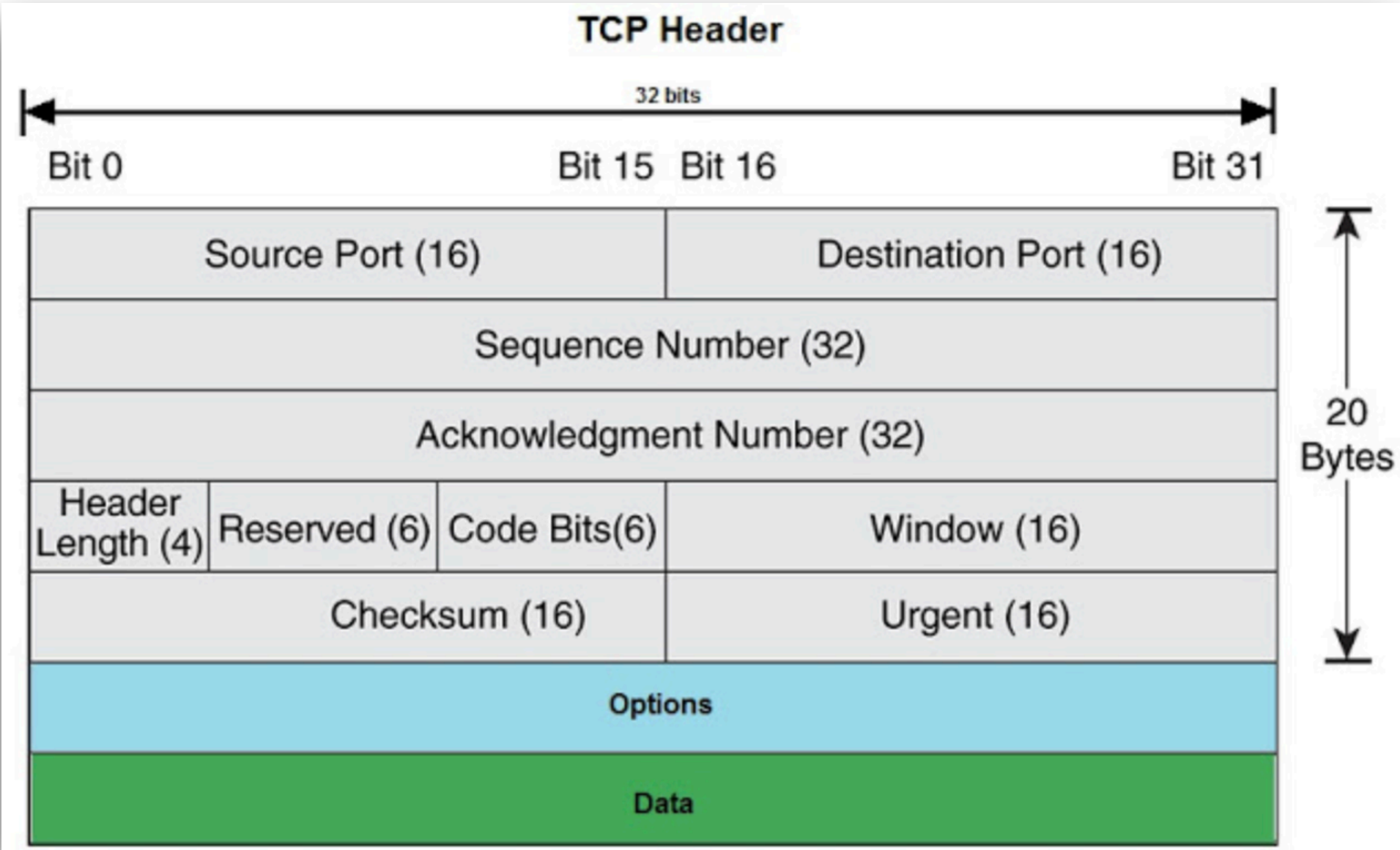
Fig. 1. Generation Framework Overview.

IPV4 Packet

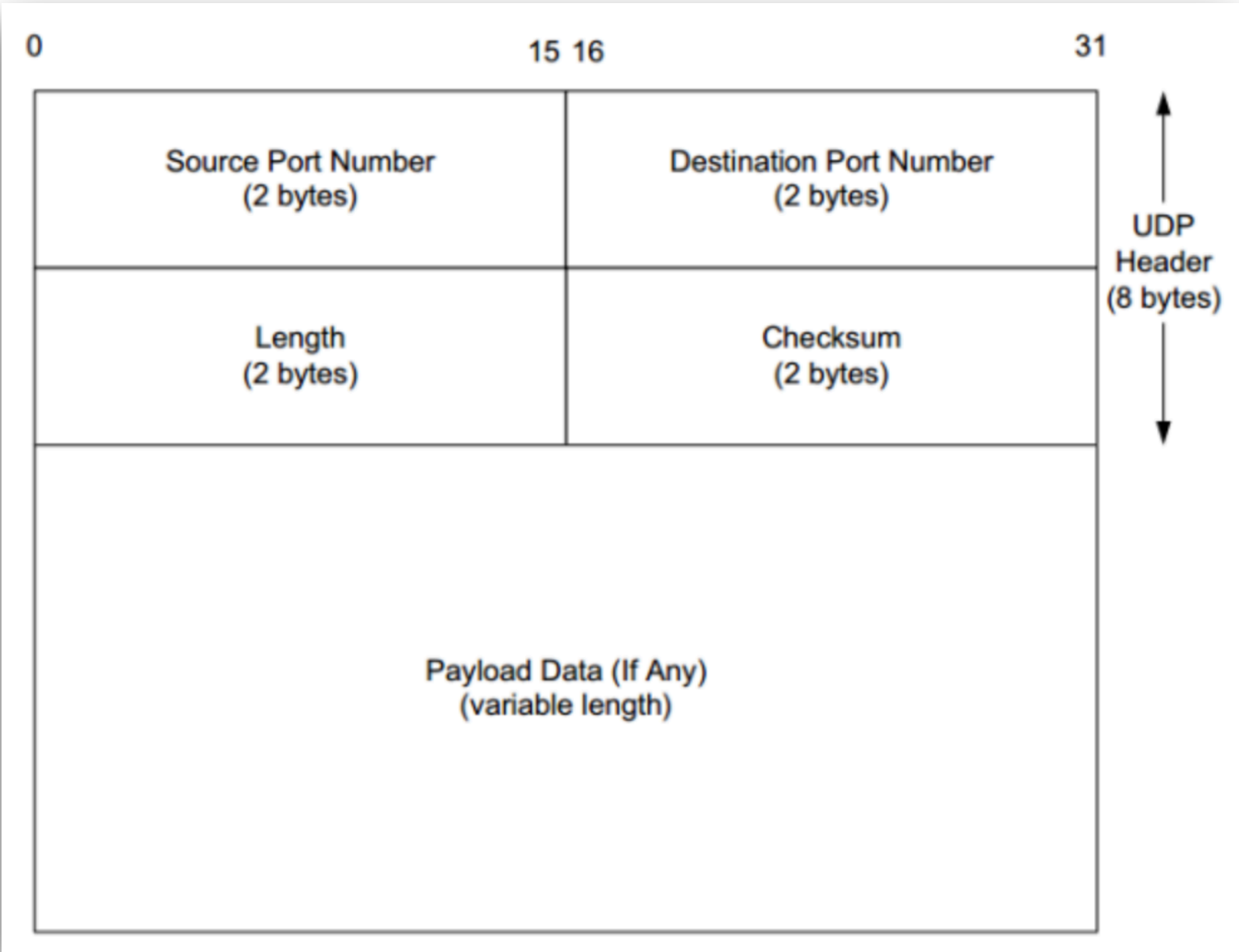


This field tells whether its Data Part will have TCP/UDP/ICMP/IGMP Style Packet

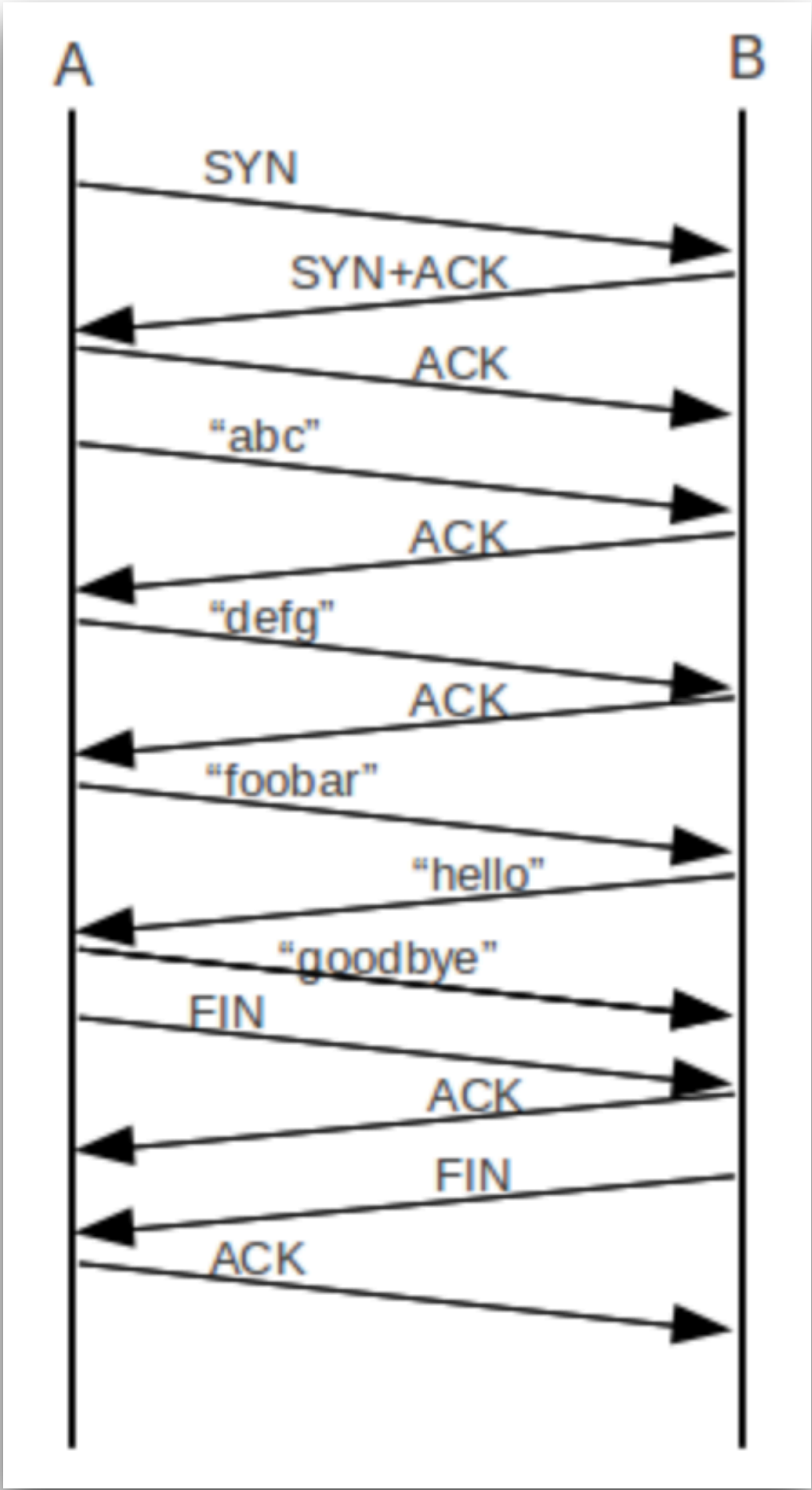
TCP Packet



UDP Packet



Typical Network Communication happening between A and B Communicating on WhatsApp



	A sends	B sends
1	SYN, seq=0	
2		SYN+ACK, seq=0, ack=1 (expecting)
3	ACK, seq=1, ack=1 (ACK of SYN)	
4	"abc", seq=1, ack=1	
5		ACK, seq=1, ack=4
6	"defg", seq=4, ack=1	
7		seq=1, ack=8
8	"foobar", seq=8, ack=1	
9		seq=1, ack=14, "hello"
10	seq=14, ack=6, "goodbye"	
11	seq=21, ack=6, FIN	seq=6, ack=21 ;; ACK of "goodbye"
12		seq=6, ack=22 ;; ACK of FIN
13		seq=6, ack=22, FIN
14	seq=22, ack=7 ;; ACK of FIN	

nPrint Header Format Illustration

Header

nPrint format

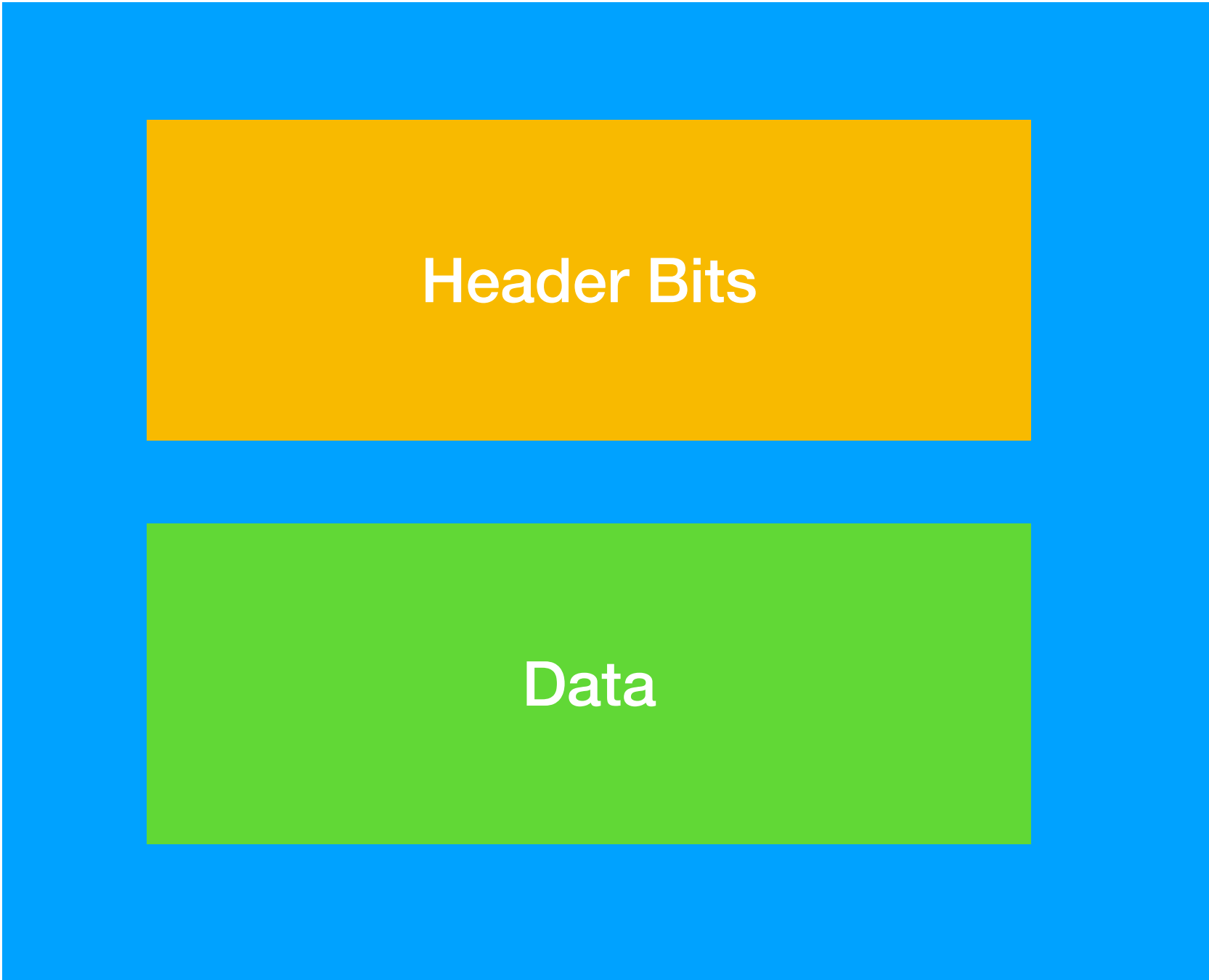
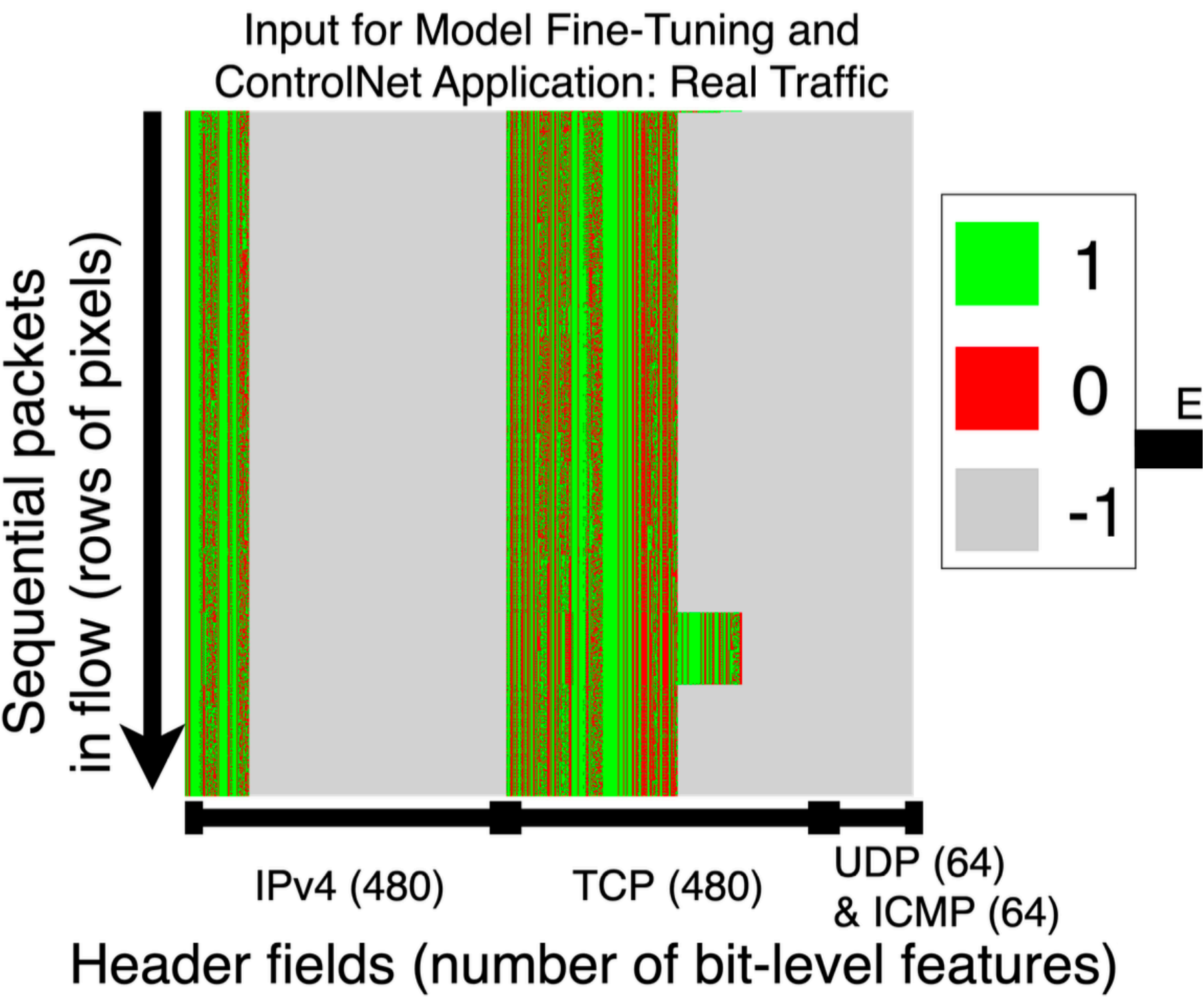


Image
encoding
→
1,0,-1

Image



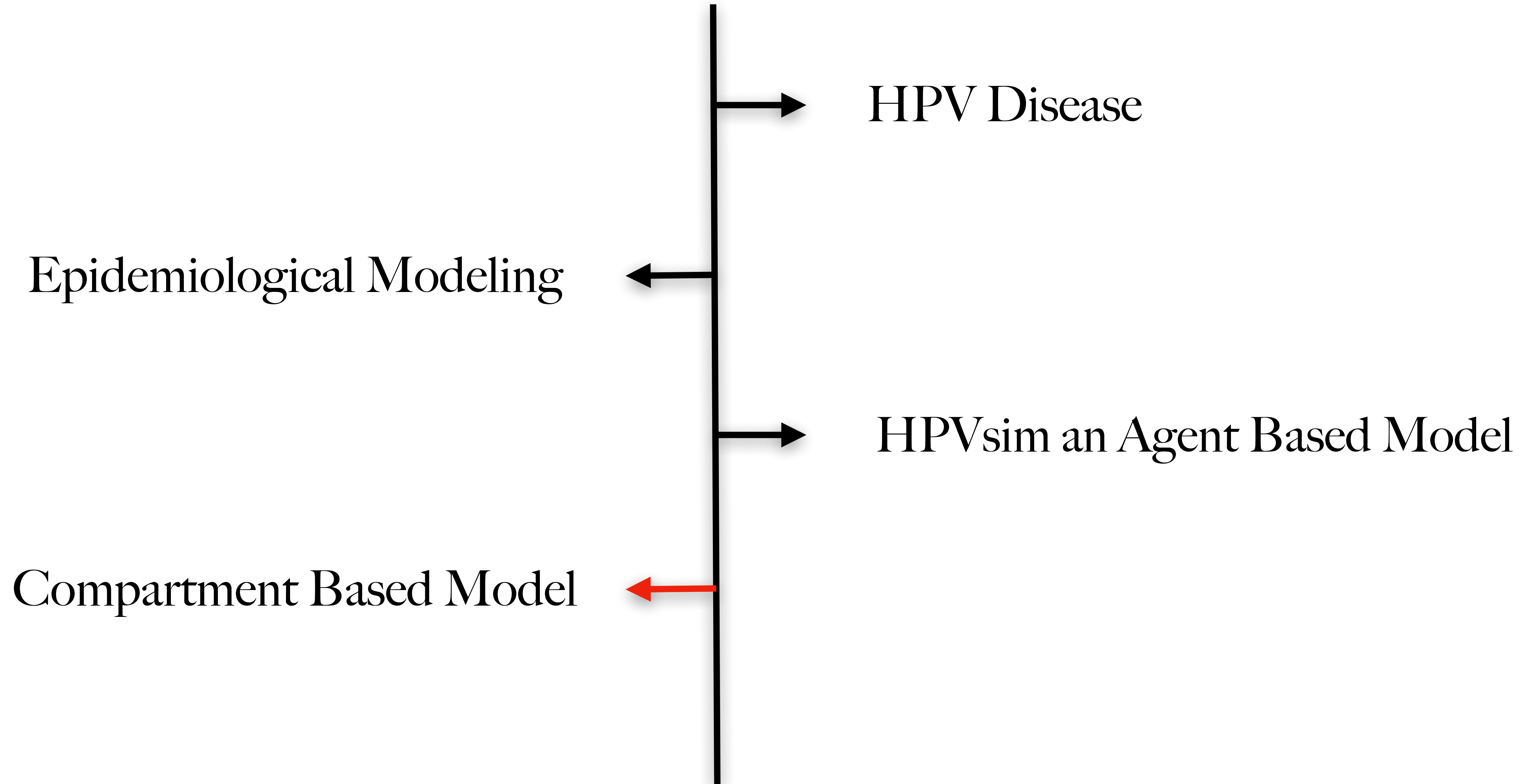
N Such Packets form N Rows in the Image

Encoding a WhatsApp chat happening between person A & person B

Improving Transport and Network Layer Protocol Compliance

Section	Details
Controlled Diffusion Model Generation	• <i>Using controlled diffusion models to generate encoded network traffic resembling real-world protocol and header field value distributions.</i> • <i>Capturing every feature observed in real network traffic in the encoded format. Minimising statistical disparity between real and synthetic feature values for improved model recognition of underlying patterns.</i>
Post-Processing Heuristic for Compliance	• <i>Identification of critical header fields requiring strict adherence to formatting rules. Limiting the scope of fields for modification during post-processing to retain the original generative model's output.</i> • <i>Construction of dependency trees for intra-packet and inter-packet dependencies based on domain knowledge and standard protocol documentation. Automated bottom-up traversal of dependency trees to correct intra-packet and inter-packet dependencies, ensuring internal consistency and correct relationships between packets.</i> • <i>Majority voting system for determining appropriate values for fields requiring uniformity across packets within the same network traffic trace. Addressing challenges like timestamp assignment using sampled time distributions from real traffic as an interim solution, with plans for future improvements.</i> • <i>Post-processing ensures that the synthetic traffic closely adheres to essential protocol rules, making it suitable for conversion into raw network traffic formats and utilization in various non-ML tasks.</i>

Road map

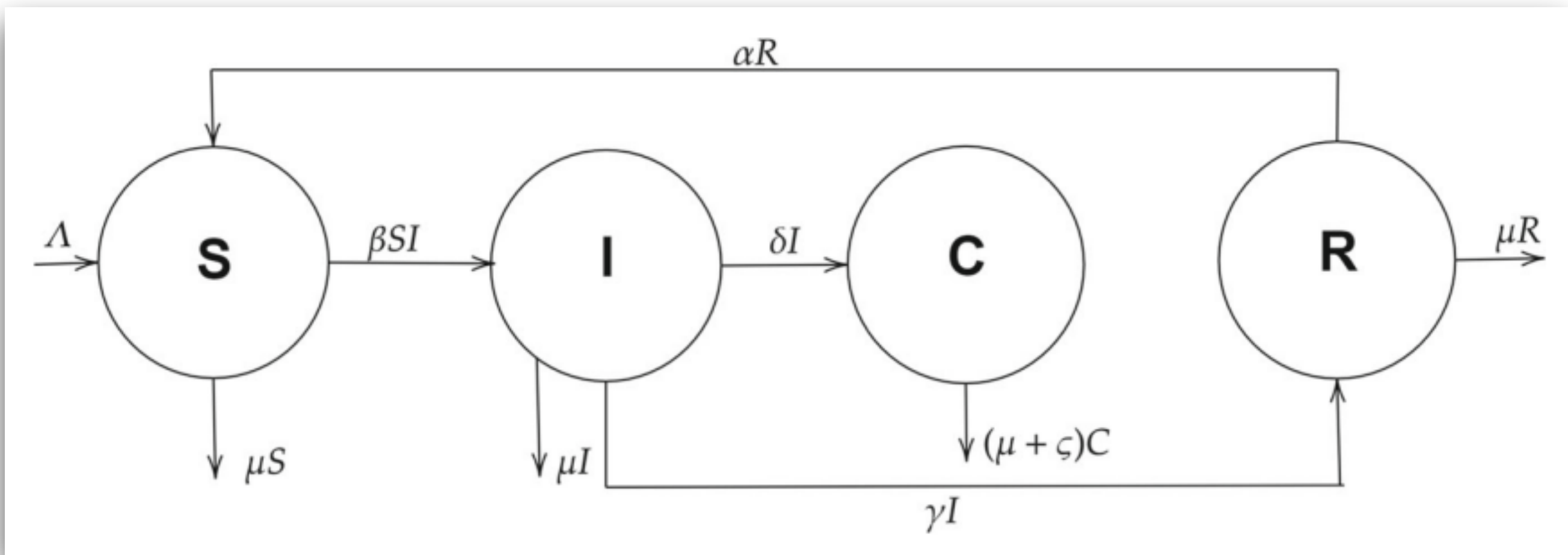


Optimal Control Theory

We use optimal control techniques with an Intention to find an Intervention schedule for HPV complying multiple constraints including Budget and Human Possibility

Model 1

Basic SICR Model

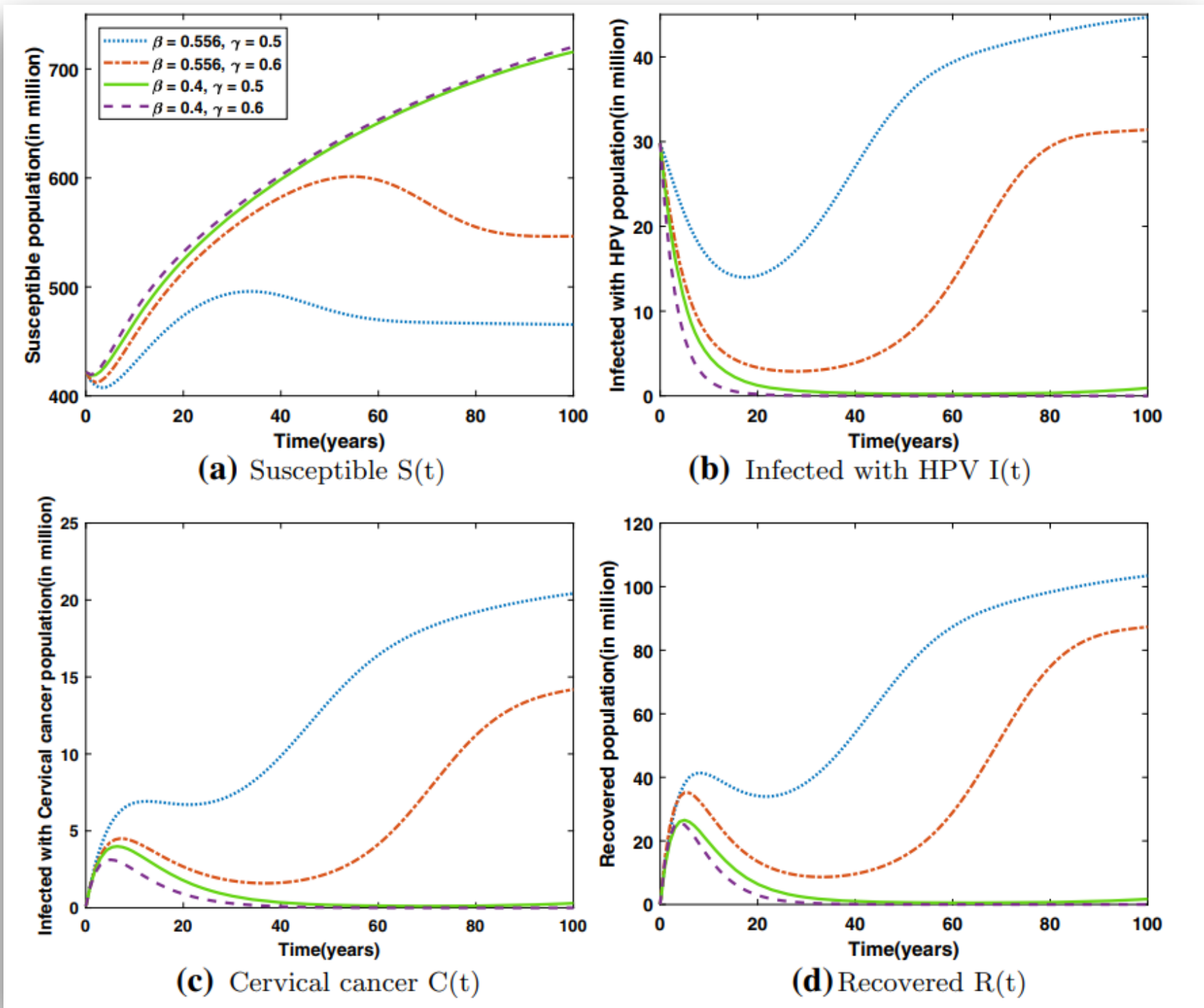


Parameter	Description	Values (year) ⁻¹
Λ	Recruitment rate into the susceptible individual	9.016228
μ	Natural rate of mortality	0.0141784
ς	Death rate of individuals with cervical cancer	0.108696
β	A transmission rate of contact with individuals	0.556
δ	Progression rate of HPV to cervical cancer	0.05714
γ	Recovery rate due to natural immunity	0.5
α	Rate of loss of immunity	0.2

ODEs:

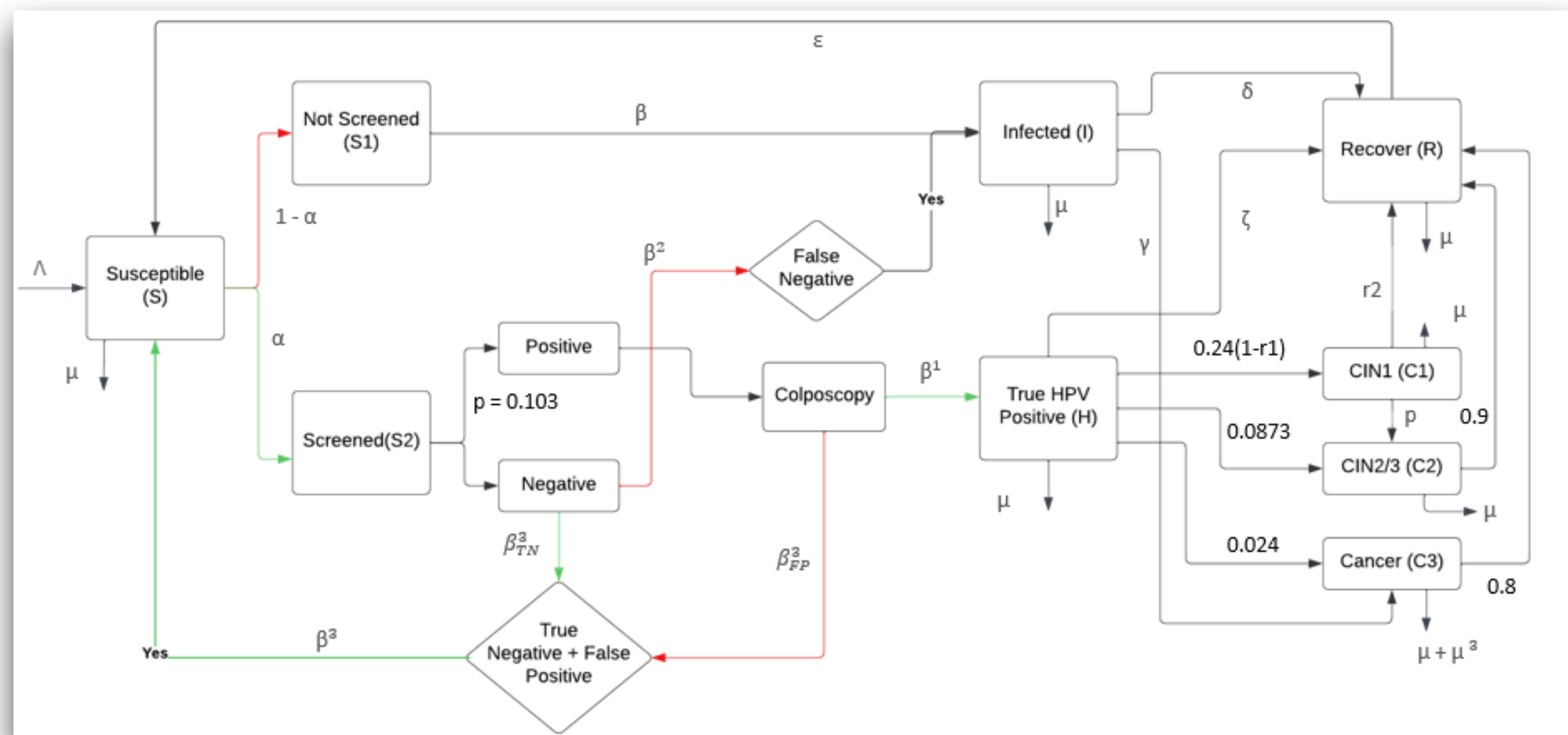
$$\begin{aligned} \frac{dS}{dt} &= \Lambda - \frac{\beta SI}{N} - \mu S + \alpha R \\ \frac{dI}{dt} &= \frac{\beta SI}{N} - (\delta + \mu + \gamma) I \\ \frac{dC}{dt} &= \delta I - (\varsigma + \mu) C \\ \frac{dR}{dt} &= \gamma I - (\alpha + \mu) R \end{aligned}$$

$$S(0) = 421.896 \text{ million}, I(0) = 29.819 \text{ million}, C(0) = 0.0991 \text{ million}, R(0) = 0$$



Model 2

Compartmental Model for Screening



- The initial susceptible population is now divided into two groups: Screened and Not Screened
- Not Screened Population: Follows the original model's path, transitioning between Infected and Recovered compartments.
- Screened Population:
 - Positive: Requires further investigation (Colposcopy).
 - True HPV Positive: Categorized based on colposcopy results (CIN 1, CIN 2/3, Cancer).
 - CIN 2/3, and Cancer patients undergo thermal ablation treatment (WHO recommendation).
 - Treatment efficacy determines progression or recovery within these compartments.
 - False Positive: Individuals mistakenly identified as positive. These individuals return to the Susceptible compartment.
 - Negative: No further action is needed at this stage, remains in the Screened compartment.
 - False Negatives: Individuals who receive a negative screening result despite being infected.
 - These individuals follow the same path as the Not Screened population.
- Late-Stage Diagnosis: A portion of the Infected population from the Not Screened group may progress directly to Cancer, reflecting potential late-stage diagnoses.

Parameters	Definition	Value
Λ	recruitment rate of susceptible women	9.016 Mil
α	Screening rate	0.7
β	HPV transmission rate	0.556
γ	rate at HPV progress to develop cervical cancer	0.485484
δ	rate of HPV recovering without cancer	0.5
μ	population's normal death rate	0.014516
μ_3	mortality rate of population after cancer	0.108696
ϵ	rate of becoming susceptible again	0.2
r_1	rate of clearance from CIN1 but remaining HPV infected	0.13
r_2	rate of clearance from both the CIN1 and infection	0.77
p	rate of progression from CIN1 to CIN2	0.065
β^1	True positive rate from Colposcopy	0.4
β^2	False negative rate during screening	0.013
β^3_{FP}	False positive rate from Colposcopy	0.6
β^3_{TN}	True negative rate during screening	0.987
η	Rate at which not screened people become susceptible again	0.429484
ζ	Recovery rate of HPV patients	0.6799

$$\beta^1 = \frac{\text{Sensitivity} \times \text{prevalence}}{\text{Sensitivity} \times \text{prevalence} + (1 - \text{Specificity}) \times (1 - \text{prevalence})} = 0.4$$

$$\beta^2 = \left[1 - \frac{\text{Specificity} \times (1 - \text{prevalence})}{(1 - \text{Sensitivity}) \times \text{prevalence} + (\text{Specificity}) \times (1 - \text{prevalence})} \right] = 0.013$$

$$\beta^3_{FP} = \left[1 - \frac{\text{Sensitivity} \times \text{prevalence}}{\text{Sensitivity} \times \text{prevalence} + (1 - \text{Specificity}) \times (1 - \text{prevalence})} \right] = 0.6$$

$$\beta^3_{TN} = \frac{\text{Specificity} \times (1 - \text{prevalence})}{(1 - \text{Sensitivity}) \times \text{prevalence} + (\text{Specificity}) \times (1 - \text{prevalence})} = 0.987$$

Prevalence = 10%

HPV Test – Sensitivity: 0.9, Specificity: 0.85

Compartmental Model for Screening

ODEs for Model

ODEs:

$$\frac{dS}{dt} = \Lambda + \varepsilon R - S + \eta S1 + \beta_{TN}^3 N + \beta_{FP}^3 Col - \mu S$$

$$\frac{dS1}{dt} = (1 - \alpha)S - (\eta)S1 - \frac{\beta S1 I}{N}$$

$$\frac{dS2}{dt} = \alpha S - S2$$

$$\frac{dI}{dt} = \frac{\beta S1 I}{N} + \beta^2 N - (\delta + \mu + \Upsilon)I$$

$$\frac{dCol}{dt} = pS2 - (\beta^1 + \beta_{FP}^3)Col$$

$$\frac{dN}{dt} = nS2 - (\beta^2 + \beta_{TN}^3)N$$

$$\frac{dH}{dt} = \beta^1 Col - \left(0.24(1 - r_1) + 0.0873 + 0.024 + \zeta + \mu\right)H$$

$$\frac{dC1}{dt} = 0.24(1 - r_1)H - (r_2 + p + \mu)C1$$

$$\frac{dC2}{dt} = 0.0873H + pC1 - 0.9C2 - \mu C2$$

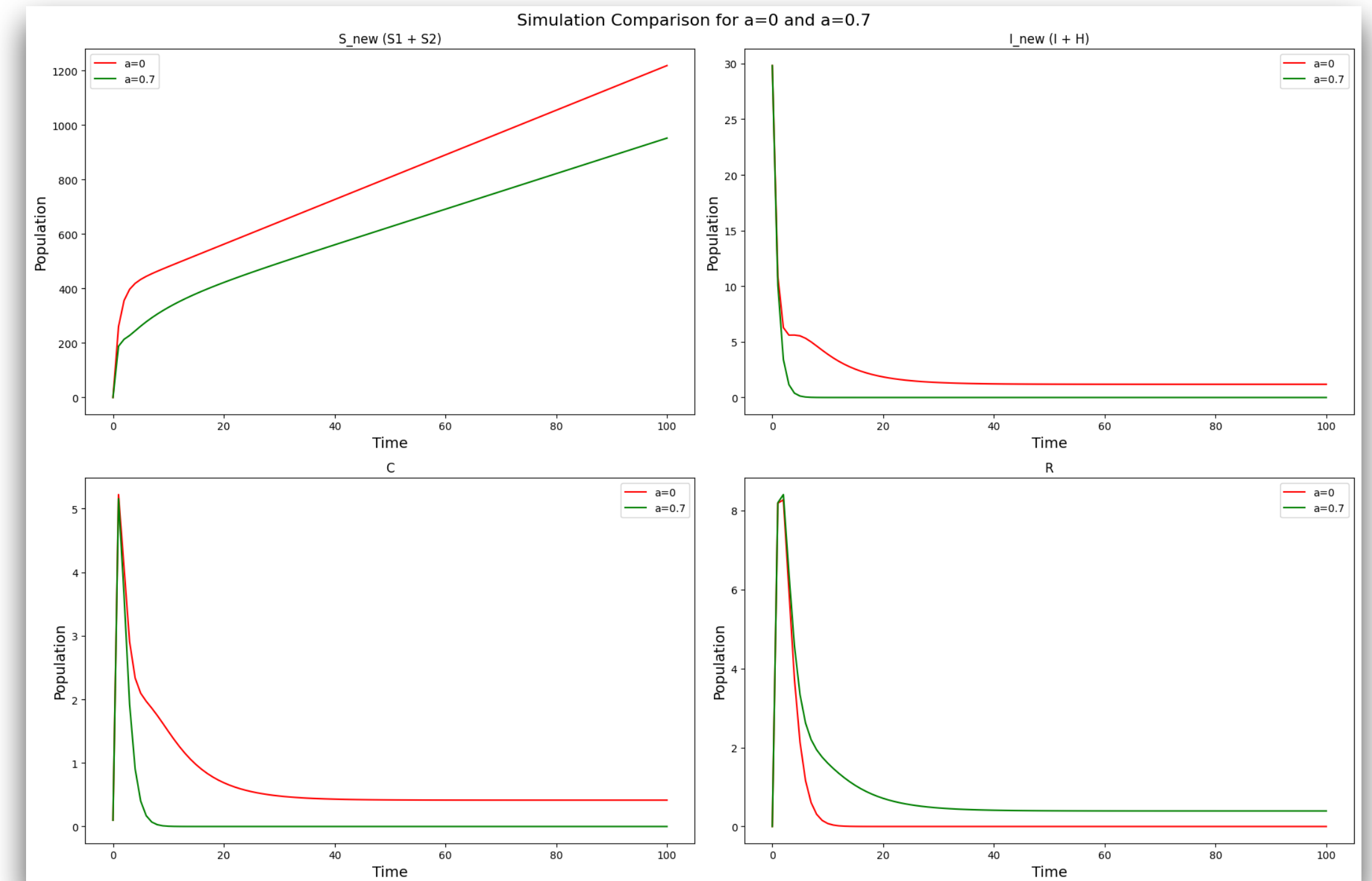
$$\frac{dC3}{dt} = 0.024H - 0.8C3 - (\mu + \mu^3)C3$$

$$\frac{dR}{dt} = \delta I + \zeta R + r_2 C1 + 0.62C2 + 0.8C3 - (\varepsilon + \mu)R$$

Compartmental Model for Screening

Simulation Results:

- **Decreased Susceptible Population (S):** Higher screening rates (α) lead to a significant decrease in the susceptible pool. This indicates successful identification and removal of individuals from the early stages of HPV infection, reducing the virus's spread.
- **Reduced Infected Population (I):** Initially, the number of infected individuals (I) also declines with increased screening. This suggests early detection through screening allows for timely interventions, potentially aiding in clearing the infection.
- **Lower Cancer Rates (C):** The model predicts a significant decrease in the cancer compartment (C) with increasing screening rates, especially in the long term. This highlights the effectiveness of screening in preventing the progression of HPV infection to cervical cancer.
- **Increased Recovered Population (R):** Initially, an increase in recovered individuals (R) is expected with higher screening. This reflects successful treatment or natural clearance of the infection after identification through screening.



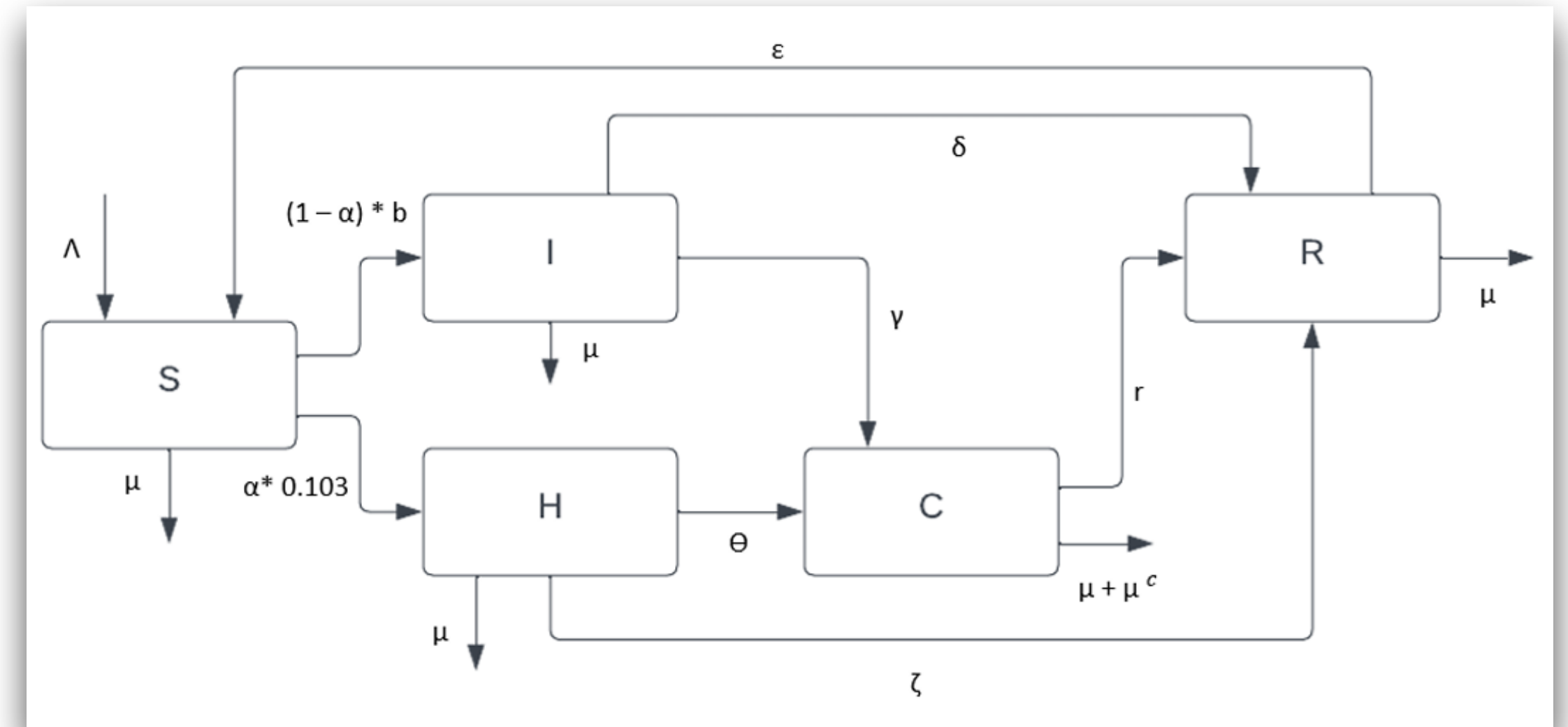
* The screening data we used is localised (497 villages in Osmanabad district, Maharashtra). But as we know cervical cancer incidence significantly varies across the country, having a nationwide randomized screening data across primary health care centres would be much more robust.

Model 3 Simplified Screening Model with Optimal Control

- Optimal control theory provides a framework for finding the best course of action (control) for a system over time, considering a desired outcome (objective) under any limitations (constraints).
- In the context of cervical cancer screening:
 - System: The population susceptible to HPV infection
 - Controls: The screening strategy, including the rate and type of screening tests offered
 - Objective: To minimize the overall cost associated with cancer management. This cost has two components:–
 - Treatment cost: cost of treating diagnosed cancer cases
 - Screening cost: cost of implementing the chosen screening program.
- We solved this optimal control problem using Forward – Backward Sweep method along with the fourth-order Runge-Kutta method for solving ODEs

$$\min \int_0^T AC(t) + B\alpha^2 S(t) dt$$

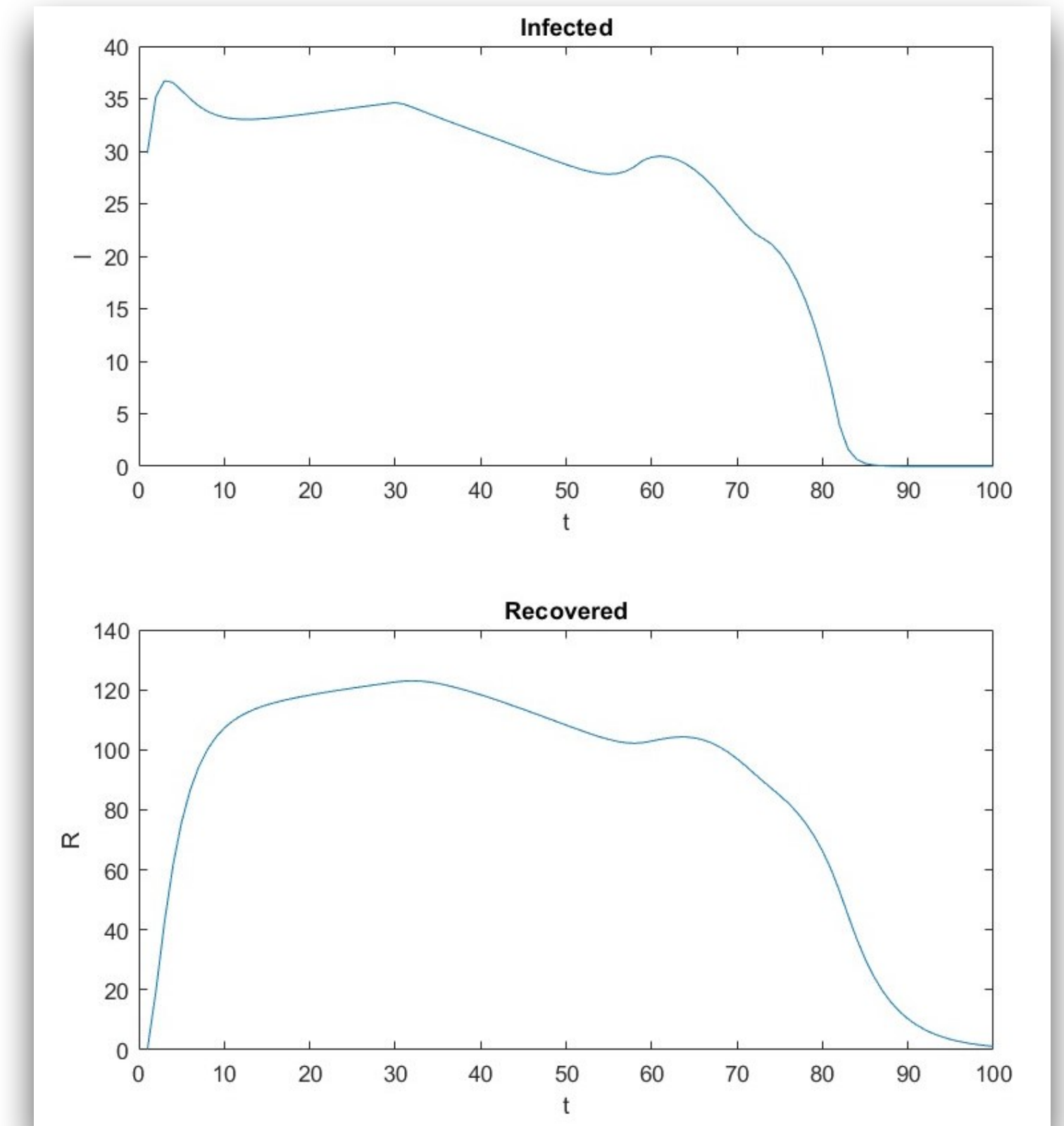
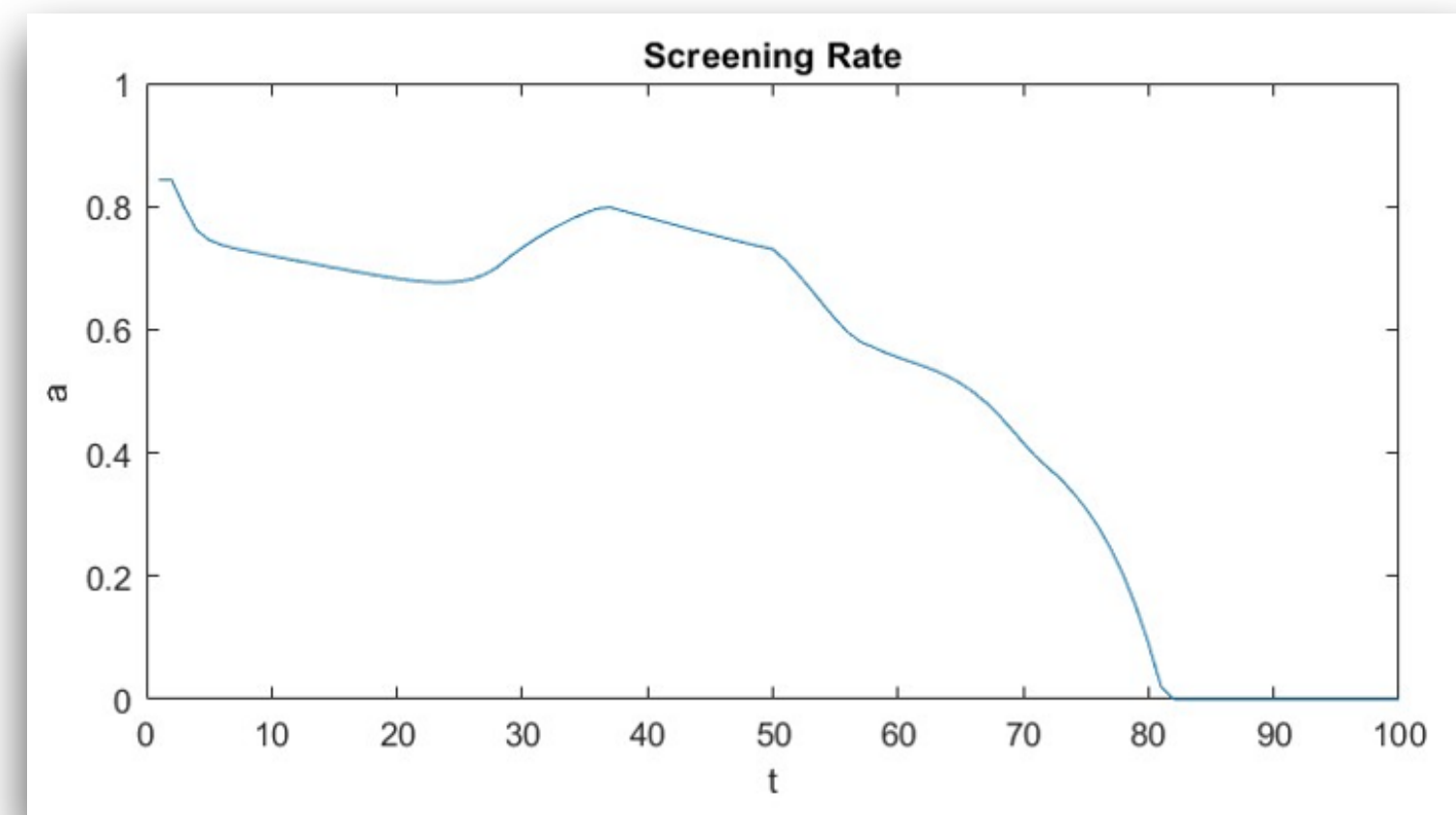
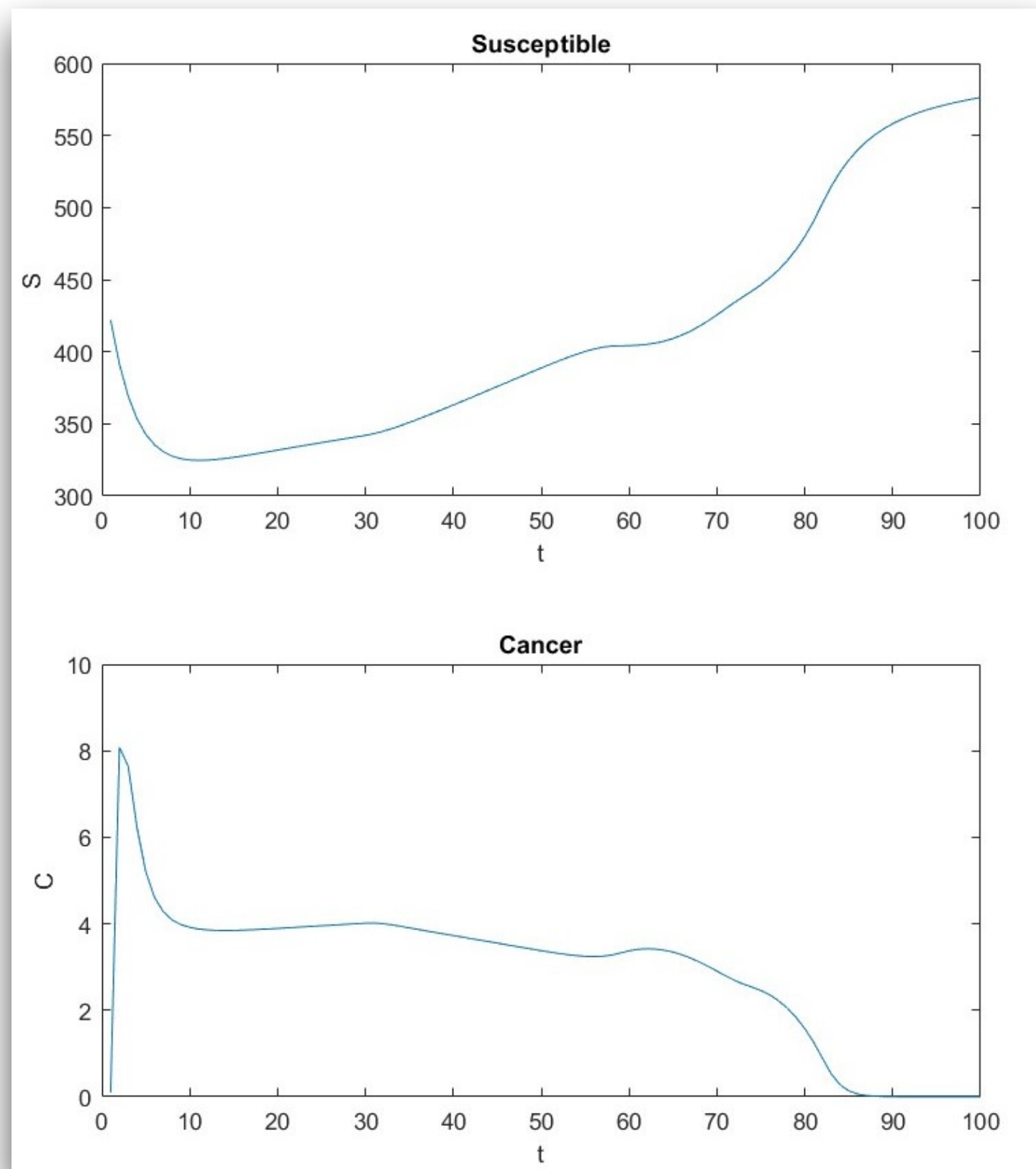
$$\begin{aligned} \frac{dS}{dt} &= \Lambda + \epsilon R - 0.103\alpha S + \frac{(1-\alpha)\beta SI}{N} - \mu S \\ \frac{dI}{dt} &= \frac{(1-\alpha)\beta SI}{N} - (\delta + \mu + \gamma)I \\ \frac{dH}{dt} &= 0.103\alpha S - (\theta + \xi + \mu)H \\ \frac{dC}{dt} &= \theta H + \gamma I - (r + \mu + \mu^2)C \\ \frac{dR}{dt} &= \delta I + \xi H + rC - (\epsilon + \mu)R \end{aligned}$$



Λ : recruitment rate of susceptible women
 α : Screening rate
 β : HPV transmission rate
 γ : rate at HPV progress to develop cervical cancer
 δ : rate of HPV recovering without cancer
 μ : population's normal death rate
 θ : rate of transmission from HPV infected to cancer
 ξ : recovery rate of HPV patients
 ϵ : rate of becoming susceptible again
 r : recovery rate from cancer
 and all the parameters are assumed to be non-negative.

Simplified Screening Model with Optimal Control

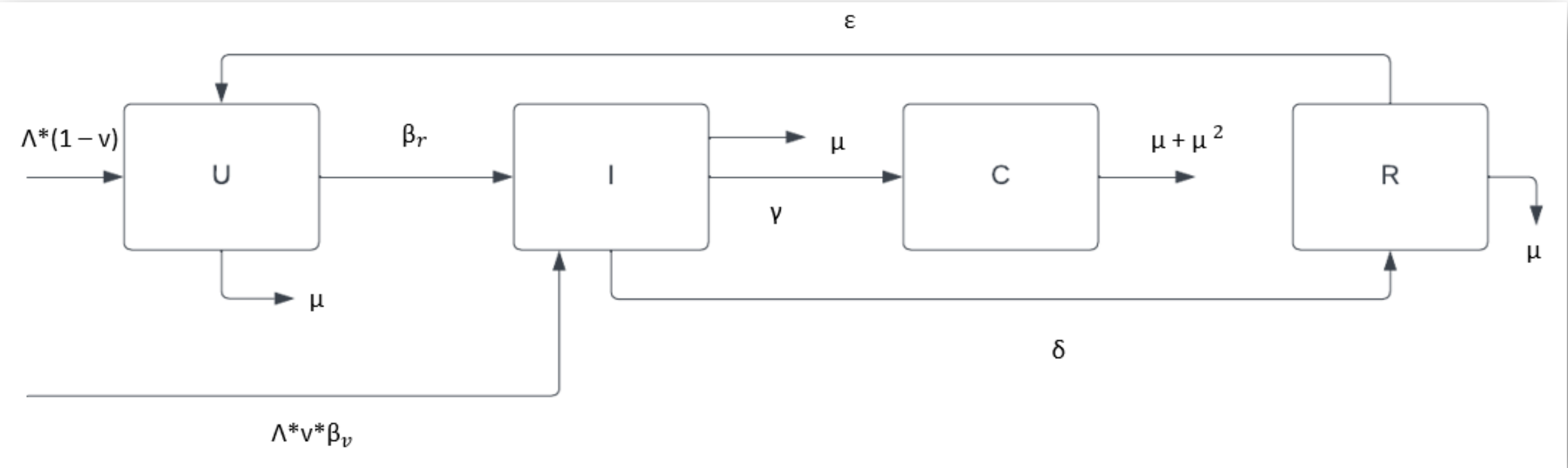
Results



Model 4

Vaccination Model

Model



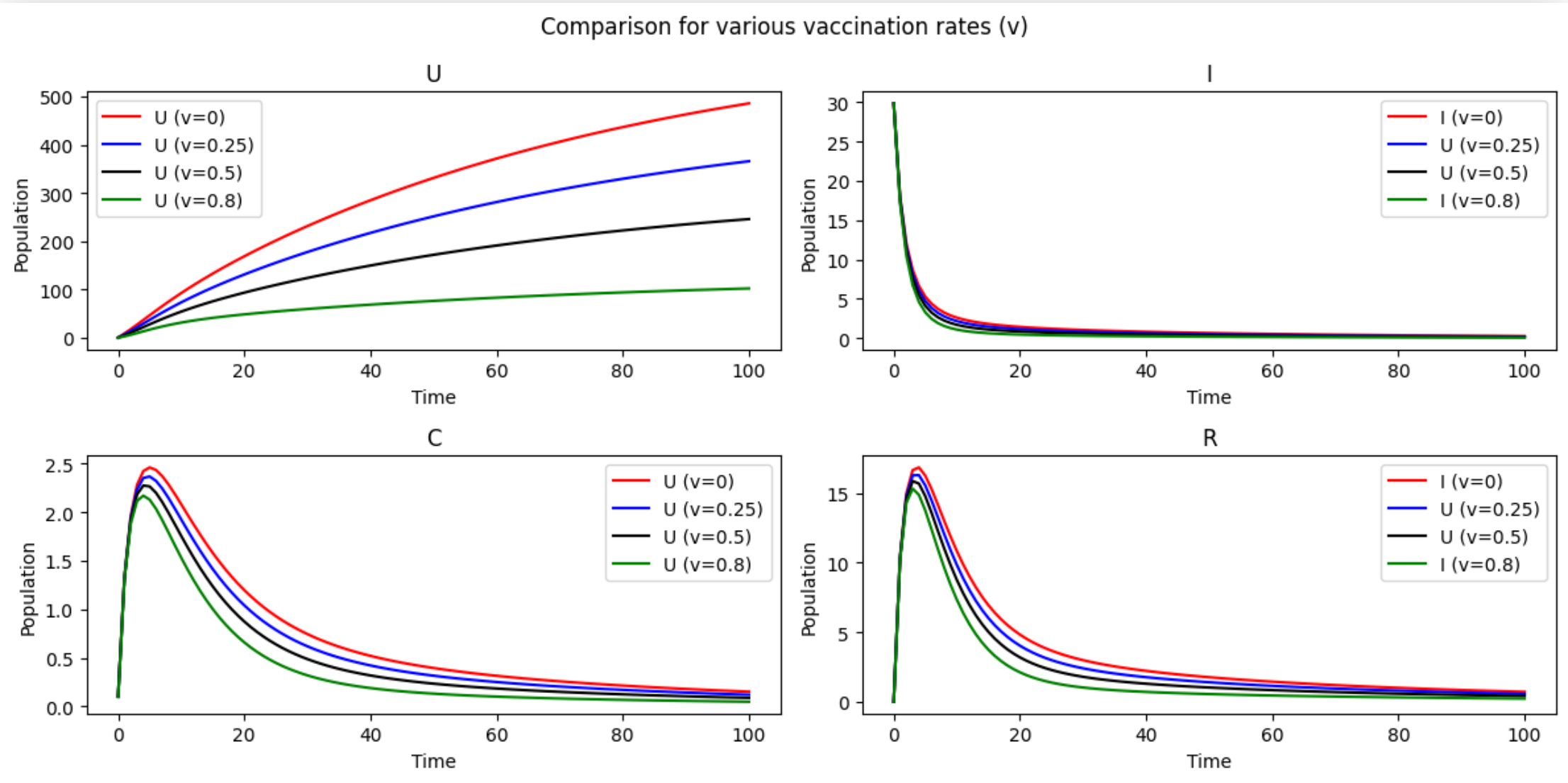
Parameters

Λ	recruitment rate of susceptible women
v	Vaccination rate
β_r	HPV transmission rate for unvaccinated women
β_v	HPV transmission rate for vaccinated women
γ	rate at HPV progress to develop cervical cancer
δ	rate of HPV recovering without cancer
μ	population's normal death rate
μ_c	mortality rate of population after cancer
ϵ	rate of becoming susceptible again

Simulation Results

ODE

$$\begin{aligned}\frac{dU}{dt} &= (1-v)\Lambda + \epsilon R - \frac{\beta_r U I}{N} - \mu U \\ \frac{dI}{dt} &= \frac{\beta_r U I}{N} + \frac{0.05 \beta_v (\Lambda v) I}{N} - (\delta + \mu + \gamma) I \\ \frac{dC}{dt} &= \gamma I - (\mu + \mu^2) C \\ \frac{dR}{dt} &= \delta I - (\mu + \epsilon) R\end{aligned}$$



Done

HPV Disease

Epidemiological Modeling

HPVsim an Agent Based Model

Compartment Based Model

